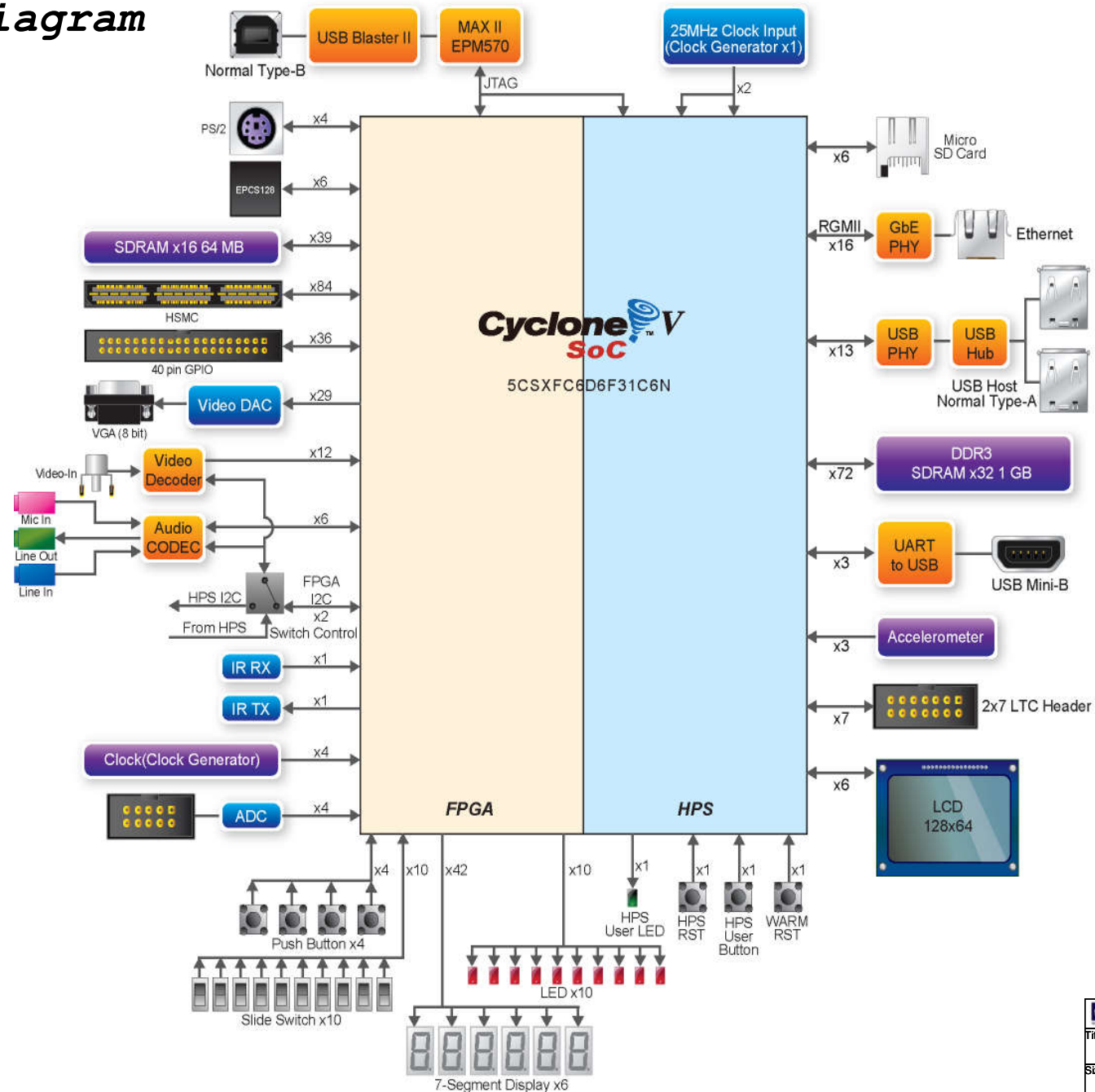


ALTERA Cyclone V SoC Development & Education Board (DE10-Standard)

PAGE	CONTENT	PAGE	CONTENT
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4	FPGA BANK 5, BANK 6	21	FPGA BUTTON, Switch
5	FPGA BANK 7, BANK 8	22	ADC, PS2, IR Tx, IR Rx
6	FPGA Clocks, GND	23	2-port USB Host
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8	FPGA Configuration	25	UART to USB, SD CARD
9	FPGA Decoupling	26	Accelerometer, LTC Connector
10	FPGA Power	27	I2C Multiplexer, HPS BUTTON, HPS LED
11	USB Blaster II	28	LCD
12	JTAG Chain	29	Power - 1.1V
13	GPIO	30	Power - 5V, 3.3V
14	HSMC	31	Power - 9V, 2.5V, 1.5V
15	SDRAM, HPS QSPI Flash	32	Power - 1.2V, 1.8V, DDR3 VREF, DDR3 VTT
16	HPS DDR3 SDRAM	33	Power - VCCIO_HSMC & HSMC_VCCPD
17	ADV7123 VGA		

Block Diagram



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Title		
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U20-9

Bank 3 VCCIO = 3.3V

Bank 3A

Bank 3B

GPIO17 AF4
GPIO18 AF5
GPIO21 AF9
GPIO16 AG8
GPIO20 AF8
GPIO15 AG7
GPIO10 AG1
GPIO6 AH2
GPIO35 AA12
GPIO34 AB12
GPIO19 AF6
GPIO14 AG6
GPIO13 AG5
GPIO9 AH5
GPIO4 AJ1
GPIO5 AJ2
GPIO33 AC12
GPIO31 AD12
GPIO11 AG2
GPIO7 AH3

IO_3A/PR_ERROR/DIFFIO_RX_B7P
IO_3A/PR_DONE/DIFFIO_RX_B7N
IO_3A/DIFFIO_TX_B8P/DQ1B
IO_3A/PR_READY/DIFFIO_TX_B8N/DQ1B
IO_3A/DIFFIO_TX_B8P/DQ2B
IO_3A/DIFFIO_TX_B9N
IO_3A/DIFFIO_RX_B10P/DQ2B
IO_3A/DIFFIO_RX_B10N/DQ2B
IO_3A/DIFFIO_RX_B11P/DQ2B
IO_3A/DIFFIO_RX_B11N/DQSN2B
IO_3A/DIFFIO_TX_B12P
IO_3A/DIFFIO_TX_B12N/DQ2B
IO_3A/DIFFIO_TX_B13P/DQ2B
IO_3A/DIFFIO_TX_B13N/DQ2B
IO_3A/DIFFIO_RX_B14P/DQ2B
IO_3A/DIFFIO_RX_B14N/DQ2B
IO_3A/DIFFIO_RX_B15N
IO_3A/DIFFIO_TX_B16P/DQ2B
IO_3A/DIFFIO_TX_B16N/DQ2B

IO_3B/DIFFIO_TX_B17P/DQ3B
IO_3B/DIFFIO_TX_B17N
IO_3B/DIFFIO_RX_B18P/DQ3B
IO_3B/DIFFIO_RX_B18N/DQ3B
IO_3B/DIFFIO_RX_B19P/DQ3B
IO_3B/DIFFIO_RX_B19N/DQSN3B
IO_3B/DIFFIO_TX_B20P
IO_3B/DIFFIO_TX_B20N/DQ3B
IO_3B/DIFFIO_TX_B21P/DQ3B
IO_3B/DIFFIO_TX_B21N/DQ3B
IO_3B/DIFFIO_RX_B22P/DQ3B
IO_3B/DIFFIO_RX_B22N/DQ3B
IO_3B/DIFFIO_RX_B23P
IO_3B/DIFFIO_RX_B23N
IO_3B/DIFFIO_TX_B24P/DQ3B
IO_3B/DIFFIO_TX_B24N/DQ3B
IO_3B/DIFFIO_TX_B25N/GND
IO_3B/DIFFIO_RX_B26P/DQ4B/B_A_14
IO_3B/DIFFIO_RX_B26N/DQ4B/B_A_15
IO_3B/DIFFIO_RX_B27P/DQSN4B/B_CSN_1
IO_3B/DIFFIO_RX_B27N/DQSN4B/B_CSN_1
IO_3B/DIFFIO_TX_B28P/B_A_12
IO_3B/DIFFIO_TX_B28N/DQ4B/B_A_13
IO_3B/DIFFIO_TX_B29P/DQ4B/B_A_10
IO_3B/DIFFIO_TX_B29N/DQ4B/B_A_11
IO_3B/DIFFIO_RX_B30P/DQ4B/B_A_8
IO_3B/DIFFIO_RX_B30N/DQ4B/B_A_9
IO_3B/DIFFIO_TX_B32P/DQ4B/B_CASN
IO_3B/DIFFIO_TX_B32N/DQ4B/B_RASN
IO_3B/DIFFIO_TX_B33P/DQ5B/B_BA_0
IO_3B/DIFFIO_TX_B33N/GND
IO_3B/DIFFIO_RX_B34P/DQ5B/B_BA_1
IO_3B/DIFFIO_RX_B34N/DQ5B/B_BA_2
IO_3B/DIFFIO_RX_B35P/DQ5B/B_BA_2
IO_3B/DIFFIO_RX_B35N/DQSN5B/B_CKN
IO_3B/DIFFIO_TX_B36P/B_A_6
IO_3B/DIFFIO_TX_B36N/DQ5B/B_A_7
IO_3B/DIFFIO_RX_B38P/DQ5B/B_A_4
IO_3B/DIFFIO_RX_B38N/DQ5B/B_A_5
IO_3B/DIFFIO_TX_B40P/DQ5B/B_A_0
IO_3B/DIFFIO_TX_B40N/DQ5B/B_A_1

AG10 DRAM_DQ5
AH9 DRAM_DQ11
AF11 15 DRAM_CAS_N
AG11 15 DRAM_CS_N
AA13 15 DRAM_WE_N
AB13 15 DRAM_LDQM
AK2 GPIO1
AK3 GPIO3
AJ4 KEY0
AK4 KEY1
AE13 15 DRAM_RAS_N
AF13 15 DRAM_BA0
AD14 DRAM_ADDR6
AE14 DRAM_ADDR3
AJ5 DRAM_DQ15
AK6 DRAM_DQ0
AJ6 DRAM_DQ14
AJ7 DRAM_DQ1
AG12 DRAM_ADDR10
AG13 DRAM_ADDR9
AB15 DRAM_ADDR4
AC14 DRAM_ADDR5
AK7 DRAM_DQ2
AK8 DRAM_DQ3
AJ9 DRAM_DQ10
AK9 DRAM_DQ4
AH13 DRAM_ADDR11
AH14 DRAM_ADDR1
AH7 DRAM_DQ13
AH8 DRAM_DQ12
AH10 DRAM_DQ8
AJ10 DRAM_DQ9
AJ11 DRAM_DQ7
AH11 DRAM_DQ6
AA14 KEY2
AA15 KEY3
AK12 15 DRAM_UDQM
AK13 15 DRAM_CKE
AG15 DRAM_ADDR2
AH15 DRAM_ADDR8
AJ14 DRAM_ADDR12
AK14 DRAM_ADDR0

U20-10

Bank 4A VCCIO = 3.3V

IO_4A/DIFFIO_TX_B41P/DQ6B/B_DQ_2
IO_4A/RZQ_0/DIFFIO_TX_B41N
IO_4A/DIFFIO_RX_B42P/DQ6B/B_DQ_1
IO_4A/DIFFIO_RX_B42N/DQ6B/B_DQ_0
IO_4A/DIFFIO_RX_B43P/DQ6B/B_DQ_5
IO_4A/DIFFIO_RX_B43N/DQSN6B/B_DQSN_0
IO_4A/DIFFIO_TX_B44P/B_ODT_0
IO_4A/DIFFIO_TX_B44N/DQ6B/B_DQ_3
IO_4A/DIFFIO_TX_B45P/DQ6B/B_DQ_6
IO_4A/DIFFIO_TX_B45N/DQ6B/B_ODT_1
IO_4A/DIFFIO_RX_B46P/DQ6B/B_DQ_5
IO_4A/DIFFIO_RX_B46N/DQ6B/B_DQ_4
IO_4A/DIFFIO_TX_B48P/DQ6B/B_DM_0
IO_4A/DIFFIO_TX_B48N/DQ6B/B_DQ_7
IO_4A/DIFFIO_TX_B49P/DQ7B/B_DQ_10
IO_4A/DIFFIO_TX_B49N/GND
IO_4A/DIFFIO_RX_B50P/DQ7B/B_DQ_9
IO_4A/DIFFIO_RX_B50N/DQ7B/B_DQ_8
IO_4A/DIFFIO_RX_B51P/DQ57B/B_DQ_1
IO_4A/DIFFIO_RX_B51N/DQSN7B/B_DQSN_1
IO_4A/DIFFIO_TX_B52P/B_CKE_1
IO_4A/DIFFIO_TX_B52N/DQ7B/B_DQ_11
IO_4A/DIFFIO_TX_B53P/DQ7B/B_DQ_14
IO_4A/DIFFIO_TX_B53N/DQ7B/B_CKE_0
IO_4A/DIFFIO_RX_B54P/DQ7B/B_DQ_13
IO_4A/DIFFIO_RX_B54N/DQ7B/B_DQ_12
IO_4A/DIFFIO_TX_B56P/DQ7B/B_DM_1
IO_4A/DIFFIO_TX_B56N/DQ7B/B_DQ_15
IO_4A/DIFFIO_TX_B57P/DQ8B/B_DQ_18
IO_4A/DIFFIO_TX_B57N/GND
IO_4A/DIFFIO_RX_B58P/DQ8B/B_DQ_17
IO_4A/DIFFIO_RX_B58N/DQ8B/B_DQ_16
IO_4A/DIFFIO_RX_B59P/DQ8B/B_DQ_2
IO_4A/DIFFIO_RX_B59N/DQSN8B/B_DQSN_2
IO_4A/DIFFIO_TX_B60P/B_RESETN
IO_4A/DIFFIO_TX_B60N/DQ8B/B_DQ_19
IO_4A/DIFFIO_TX_B61P/DQ8B/B_DQ_22
IO_4A/DIFFIO_TX_B61N/DQ8B/B_DQ_20

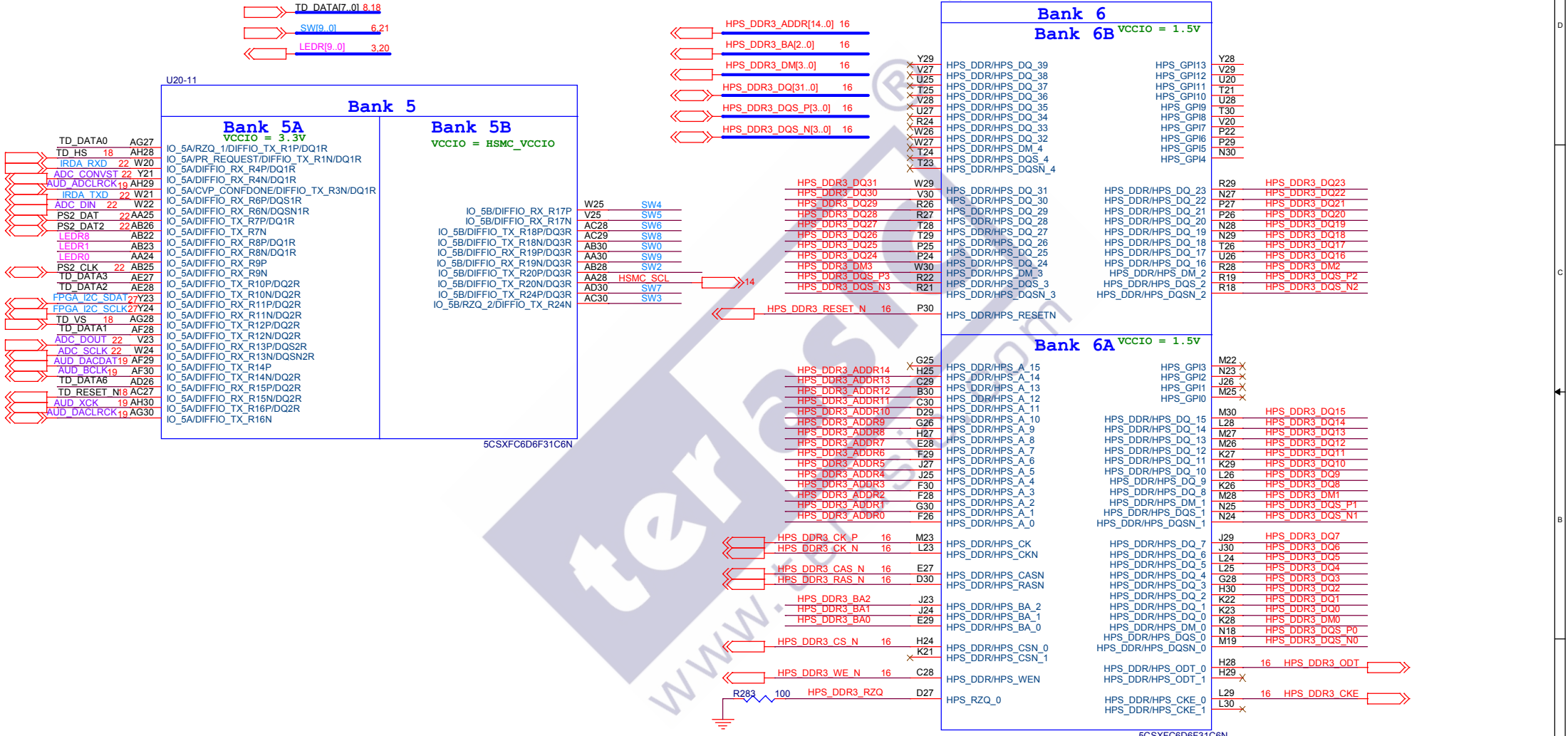
IO_4A/DIFFIO_RX_B62P/DQ8B/B_DQ_21
IO_4A/DIFFIO_RX_B62N/DQ8B/B_DQ_20
IO_4A/DIFFIO_RX_B63P/GND
IO_4A/DIFFIO_RX_B63N/GND
IO_4A/DIFFIO_TX_B64P/DQ8B/B_DM_2
IO_4A/DIFFIO_TX_B64N/DQ8B/B_DQ_23
IO_4A/DIFFIO_TX_B65P/DQ9B/B_DQ_26
IO_4A/DIFFIO_TX_B65N/GND
IO_4A/DIFFIO_RX_B66P/DQ9B/B_DQ_25
IO_4A/DIFFIO_RX_B66N/DQ9B/B_DQ_24
IO_4A/DIFFIO_RX_B67P/DQSN9B/B_DQSN_3
IO_4A/DIFFIO_RX_B67N/DQSN9B/B_DQSN_3
IO_4A/DIFFIO_TX_B68P/GND
IO_4A/DIFFIO_TX_B68N/DQ9B/B_DQ_27
IO_4A/DIFFIO_TX_B69P/DQ9B/B_DQ_30
IO_4A/DIFFIO_TX_B69N/DQ9B/B_DQ_29
IO_4A/DIFFIO_TX_B70N/DQ9B/B_DQ_28
IO_4A/DIFFIO_RX_B71P/GND
IO_4A/DIFFIO_RX_B71N/GND
IO_4A/DIFFIO_TX_B72P/DQ9B/B_DM_3
IO_4A/DIFFIO_TX_B72N/DQ9B/B_DM_3
IO_4A/DIFFIO_TX_B73N/GND
IO_4A/DIFFIO_RX_B74P/DQ10B/B_DQ_33
IO_4A/DIFFIO_RX_B74N/DQ10B/B_DQ_32
IO_4A/DIFFIO_RX_B75P/DQ10B/B_DQ_34
IO_4A/DIFFIO_RX_B75N/DQSN10B/B_DQSN_4
IO_4A/DIFFIO_TX_B76P/GND
IO_4A/DIFFIO_TX_B76N/DQ10B/B_DQ_35
IO_4A/DIFFIO_TX_B77P/DQ10B/B_DQ_38
IO_4A/DIFFIO_TX_B77N/DQ10B/B_DQ_37
IO_4A/DIFFIO_RX_B78P/DQ10B/B_DQ_37
IO_4A/DIFFIO_RX_B78N/DQ10B/B_DQ_36
IO_4A/DIFFIO_RX_B79P/GND
IO_4A/DIFFIO_RX_B79N/GND
IO_4A/DIFFIO_TX_B80P/DQ10B/B_DM_4
IO_4A/DIFFIO_TX_B80N/DQ10B/B_DQ_39

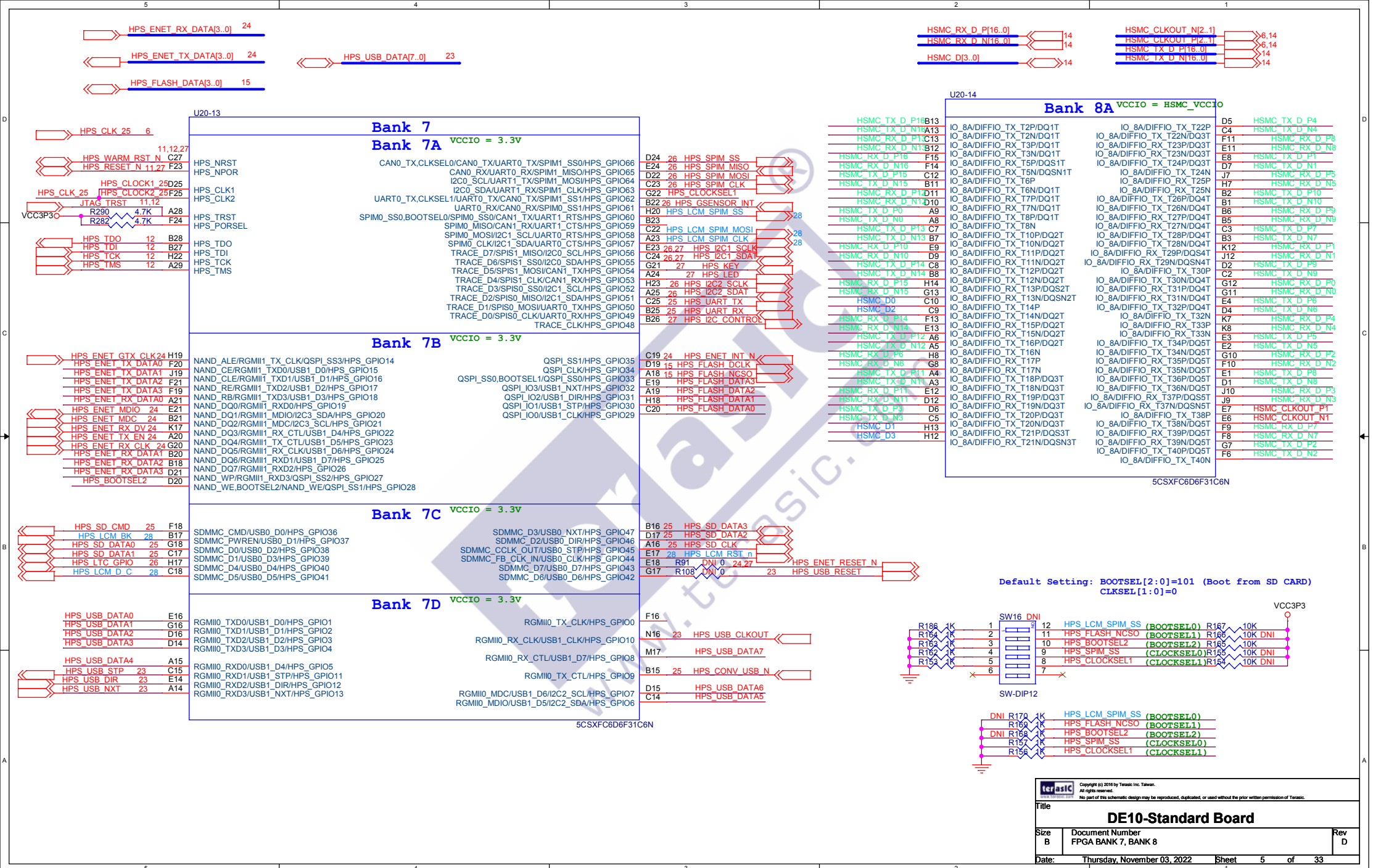
AF20 HEX52
AF21 HEX50
Y18 HEX24
AA19 HEX35
AK23 VGA_G6
AK24 VGA_G3
AJ24 VGA_G4
AJ25 VGA_G1
AF23 HEX45
AF24 LEDR7
AC20 HEX34
AD19 HEX32
AJ26 VGA_R7
AK26 VGA_G0
AG25 LEDR4
AH25 VGA_G2
AE22 HEX42
AE23 HEX43
V18 HEX01
W19 HEX31
AJ27 VGA_R3
AK27 VGA_R2
AK28 VGA_R1
AK29 VGA_R0
AD20 HEX36
AD21 HEX40
Y19 HEX30
AA20 HEX33
AG26 VGA_R6
AH27 VGA_R4
AF25 LEDR5
AF26 VGA_R5
AC22 LEDR9
AC23 LEDR2
AA21 HEX20
AB21 HEX56
AD24 LEDR3
AE24 LEDR6

5CSXFC6D6F31C6N

5CSXFC6D6F31C6N

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Clocks

Bank 3B VCCIO = 3.3V

IO_3B/CLK0P,FPLL_BL_FBP/DIFFIO_RX_B31P
IO_3B/CLK0N,FPLL_BL_FBN/DIFFIO_RX_B31N
GPIO0 W15
GPIO2 Y16
IO_3B/CLK1P/DIFFIO_RX_B39P
IO_3B/CLK1N/DIFFIO_RX_B39N

Bank 4A VCCIO = 3.3V

IO_4A/CLK2P/DIFFIO_RX_B47P
IO_4A/CLK2N/DIFFIO_RX_B47N
IO_4A/CLK3P/DIFFIO_RX_B55P
IO_4A/CLK3N/DIFFIO_RX_B55N

Bank 5B VCCIO = HSMC_VCCIO

IO_5B/CLK4P,FPLL_BR_FBP/DIFFIO_RX_R23P/DQ3R
IO_5B/CLK4N,FPLL_BR_FBN/DIFFIO_RX_R23N/DQ3R
IO_5B/CLK5P/DIFFIO_RX_R21P/DQ53R
IO_5B/CLK5N/DIFFIO_RX_R21N/DQ53R

Bank 8A VCCIO = HSMC_VCCIO

IO_8A/CLK6P,FPLL_TL_FBP/DIFFIO_RX_T9P
IO_8A/CLK6N,FPLL_TL_FBN/DIFFIO_RX_T9N
IO_8A/CLK7P/DIFFIO_RX_T1P
IO_8A/CLK7N/DIFFIO_RX_T1N

5CSXFC6D6F31C6N

CLOCK_50 AF14
DRAM_ADDR7 AF15
GPIO0 W15
GPIO2 Y16

CLOCK2_50 AA16
HEX21 AB17
TD_CLK27 18 AC18
HEX13 AD17

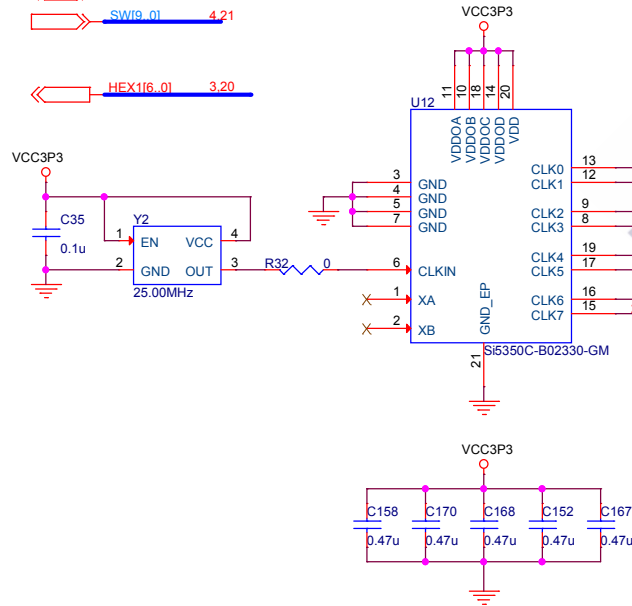
CLOCK3_50 Y26
SW1 Y27
HSMC_CLKIN_P1 AA26
HSMC_CLKIN_N1 AB27

CLOCK4_50 K14
HSMC_CLKIN0 14 J14
HSMC_CLKIN_P2 H15
HSMC_CLKIN_N2 G15

HEX216_01 3.20
DRAM_ADDR[12..0] 3.15

GPIO[35..0] 3.8,13
SW19_01 4.21

HEX116_01 3.20

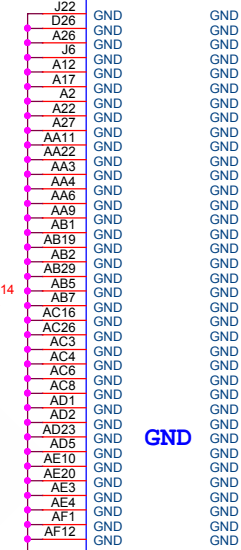


AH12 15 DRAM_CLK
AJ12 15 DRAM_BA1

AE29 HSMC_SDA
AD29 14 HSMC_CLKOUT0

A11 HSMC_CLKOUT_P2
A10 HSMC_CLKOUT_N2

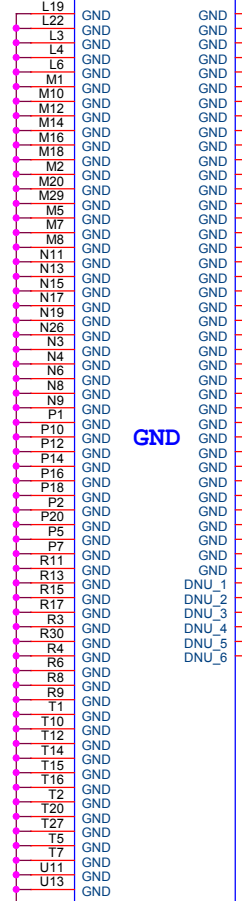
U20-6



GND

5CSXFC6D6F31C6N

U20-7

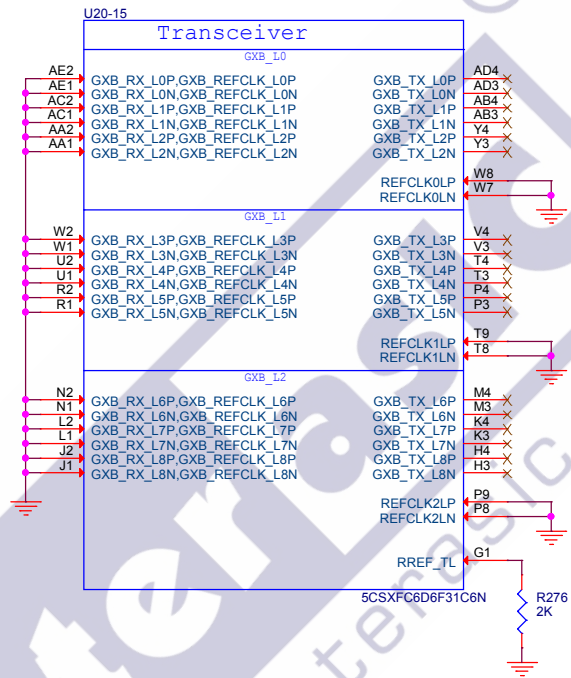


GND

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DNU_1 G2
DNU_2 AA7
DNU_3 AD15
DNU_4 E26
DNU_5 J15
DNU_6

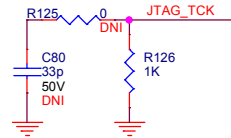
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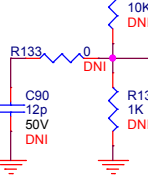
USB Blaster

FPGA TDI	12
JTAG TMS	12
JTAG TCK	12
FPGA TDO	12

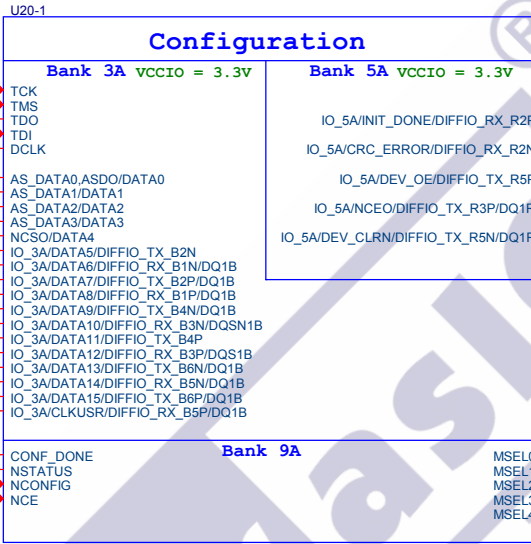


Design Note:
Optional termination resistor
for DCLK

CAD Note:
Place near FPGA DCLK pin

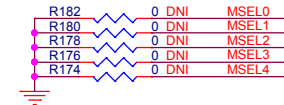
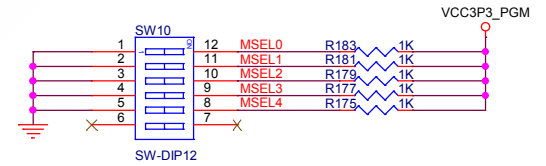


JTAG_TCK	AC5
JTAG_TMS	V9
FPGA_TDO	AB9
FPGA_TDI	U8
EPCQ_DCLK	U7
EPCQ_AS_DATA0	AE6
EPCQ_AS_DATA1	AE5
EPCQ_AS_DATA2	AE8
EPCQ_AS_DATA3	AC7
EPCQ_NCS0	AB8
GPIO24	AE9
GPIO26	AE12
GPIO28	AD9
GPIO30	AD11
GPIO22	AF10
GPIO29	AD10
GPIO25	AE11
GPIO32	AC9
GPIO8	AH4
GPIO23	AE7
GPIO12	AG3
GPIO27	AD7



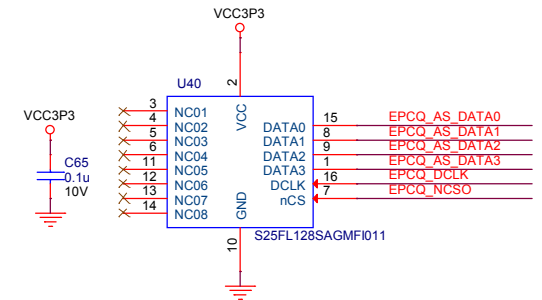
AD25	TD_DATA7
AC25	22 PS2_CLK2
AE26	TD_DATA4
AJ29	19 AUD_ADCCDAT
AD27	TD_DATA5

Fix MSEL[4:0]=10010 in AS Fast Mode



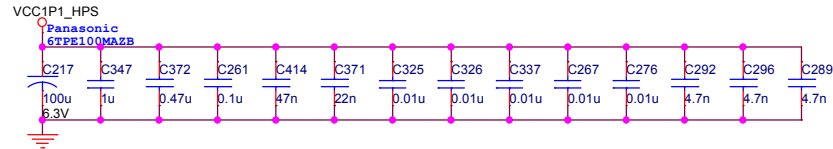
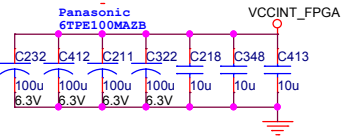
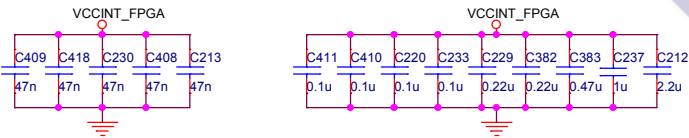
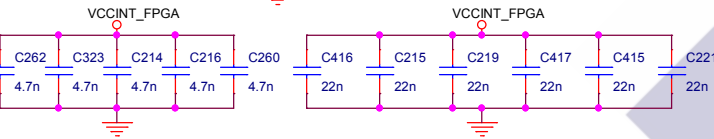
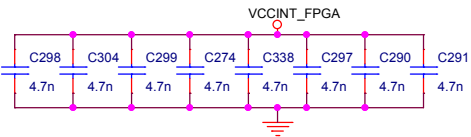
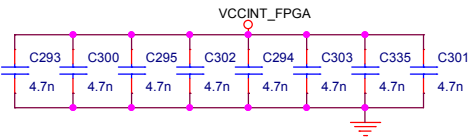
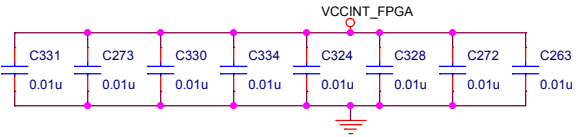
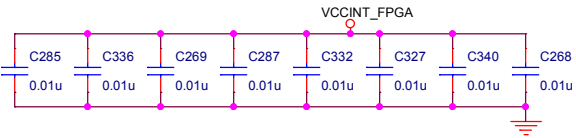
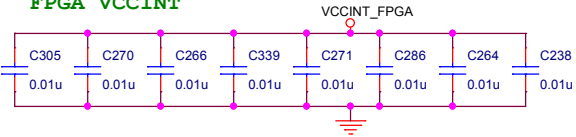
TD_DATA[7..0] 4,18

GPIO[35..0] 3,6,13

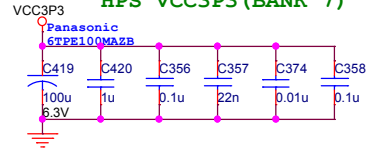


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FPGA VCCINT

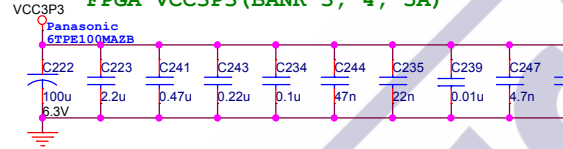


HPS VCC3P3 (BANK 7)

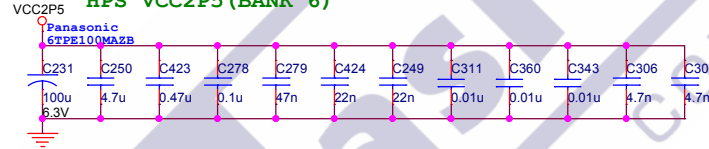


Place C394 close to J20/G23 pin

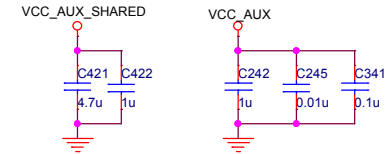
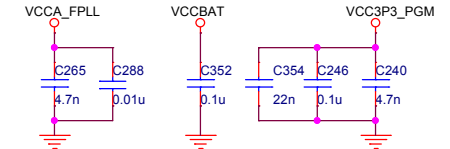
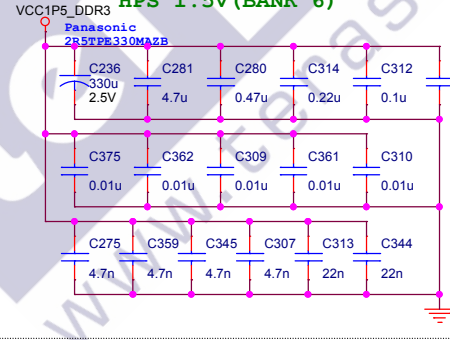
FPGA VCC3P3 (BANK 3, 4, 5A)



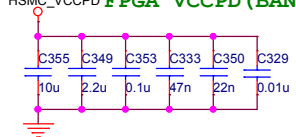
HPS VCC2P5 (BANK 6)



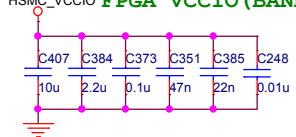
HPS 1.5V (BANK 6)

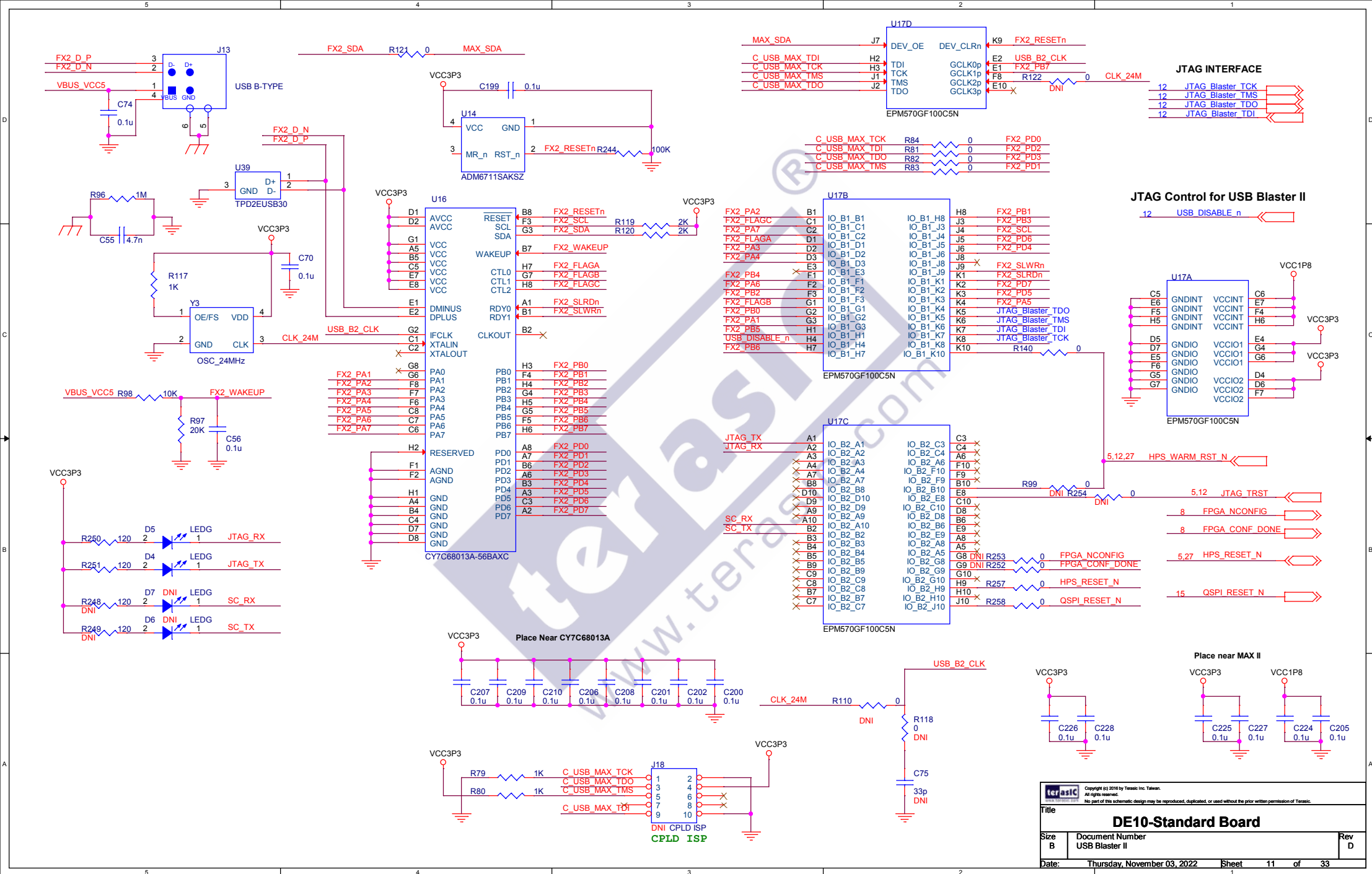


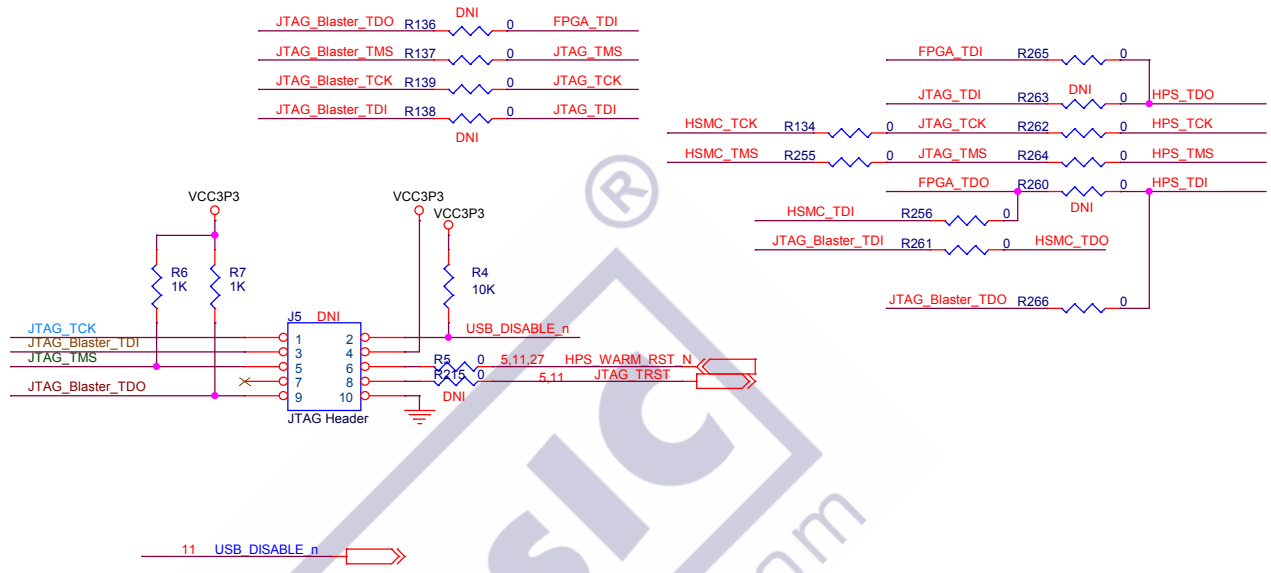
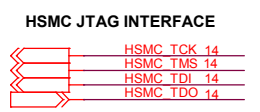
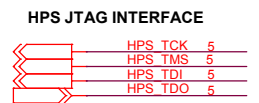
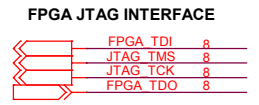
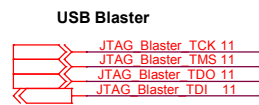
HSMC_VCCPD FPGA VCCPD (BANK 5B, 8A)



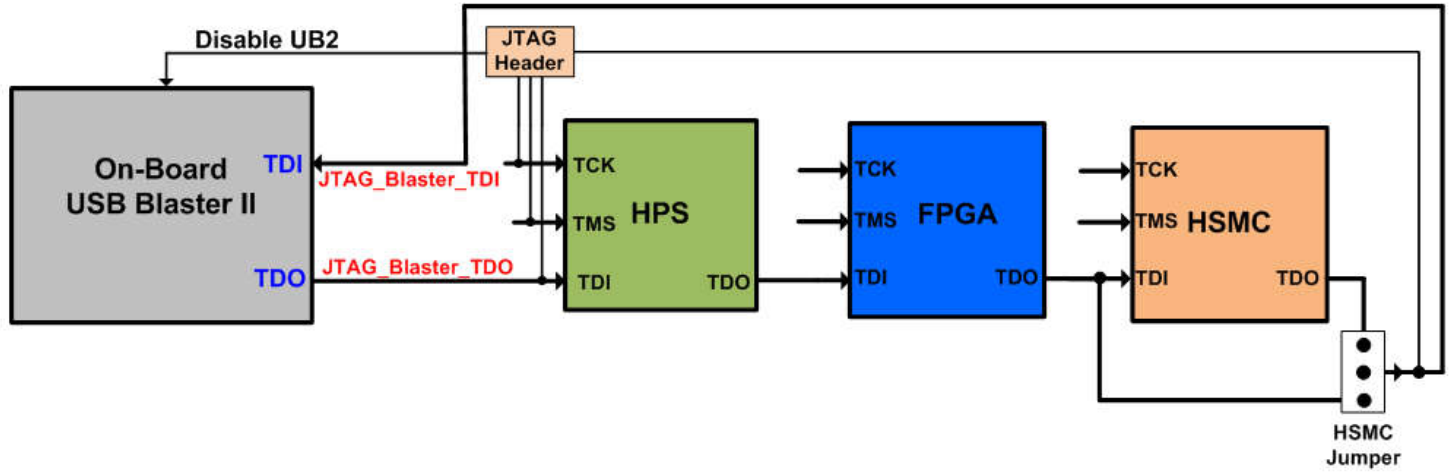
HSMC_VCCIO FPGA VCCIO (BANK 5B, 8A)

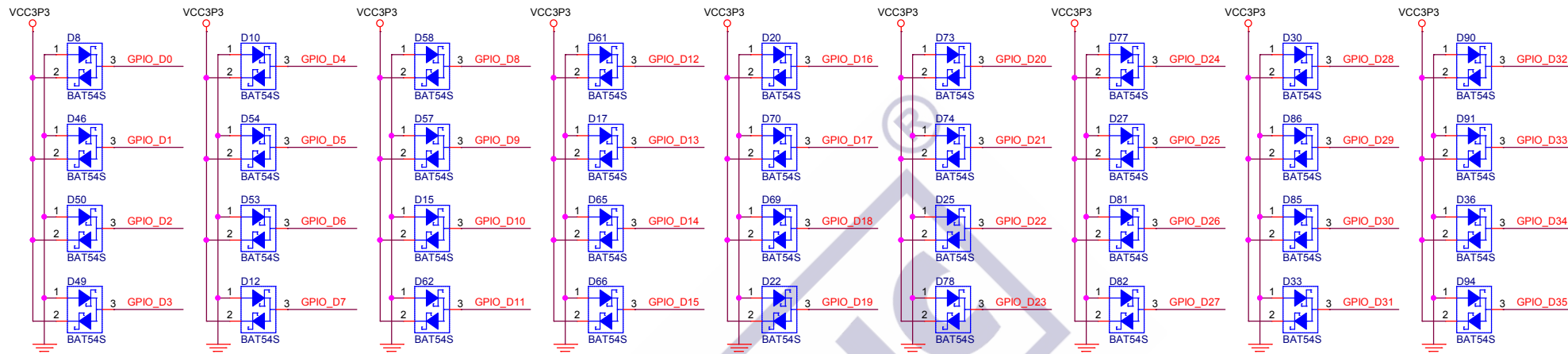




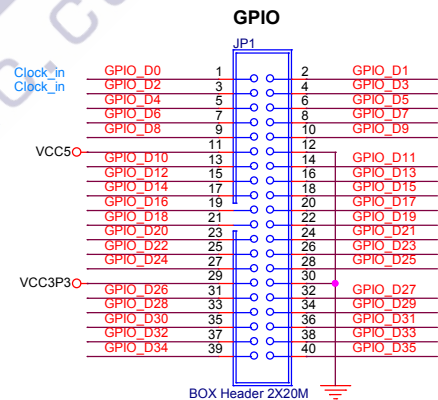
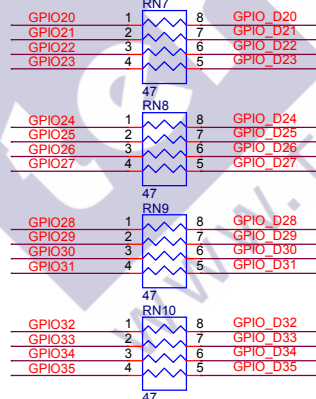
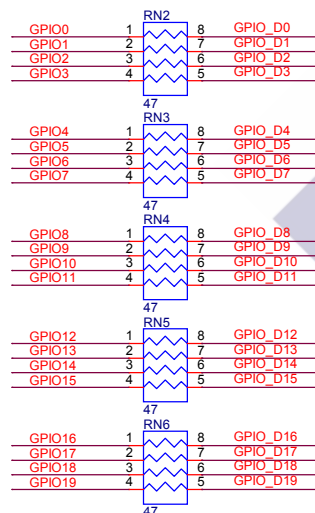


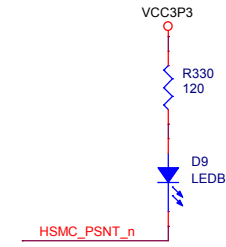
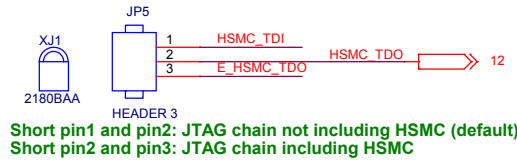
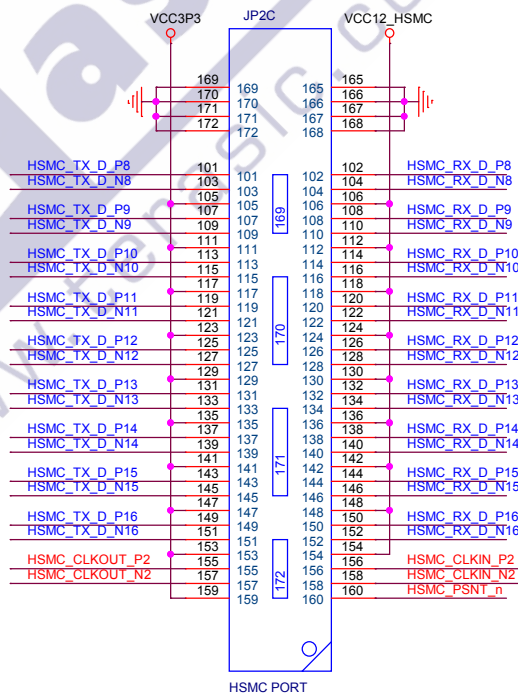
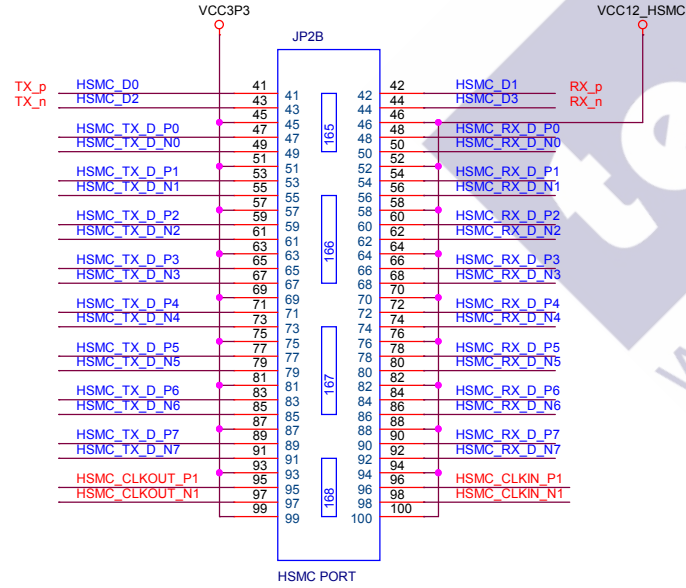
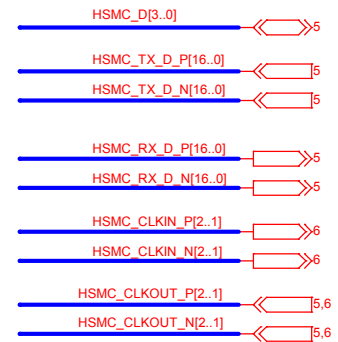
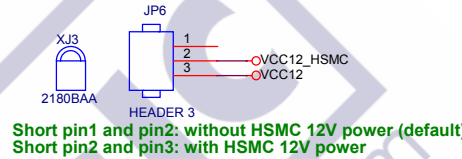
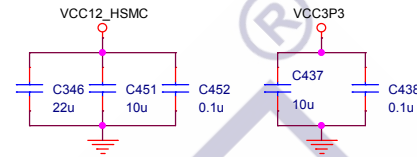
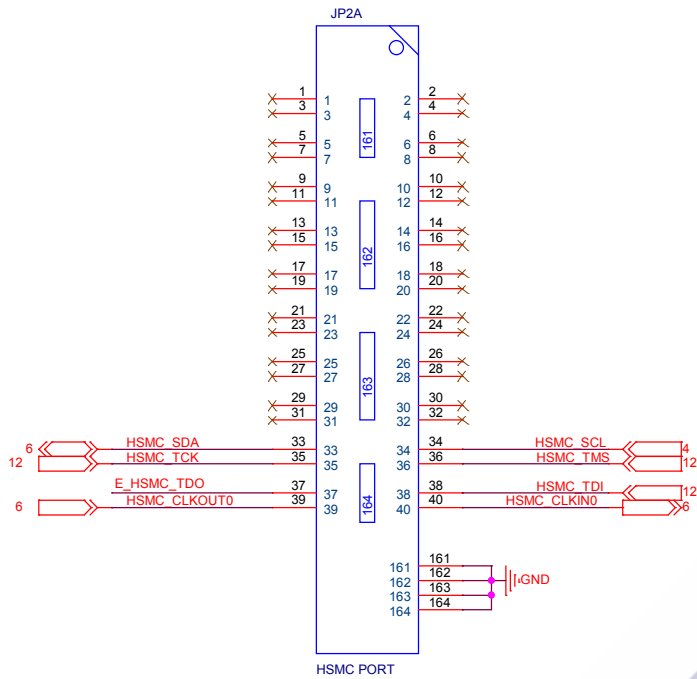
JTAG Chain



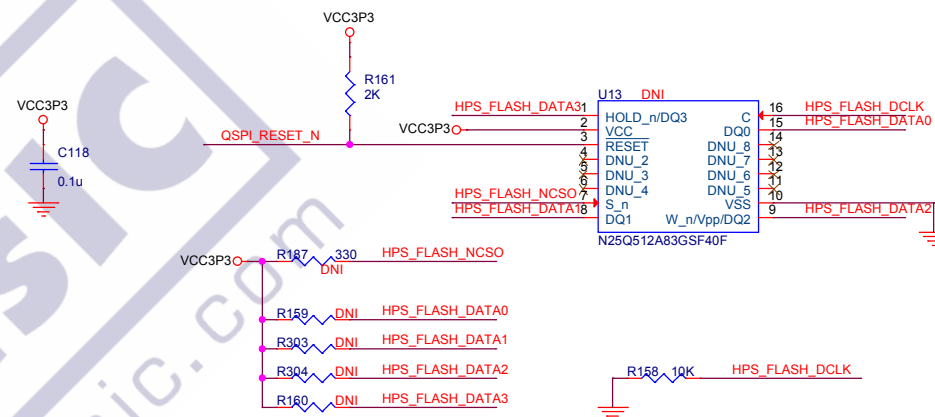
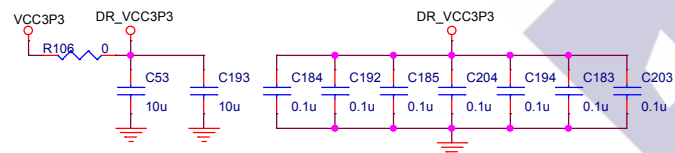


GPIO 0
GPIO[35..0] 3,6,8

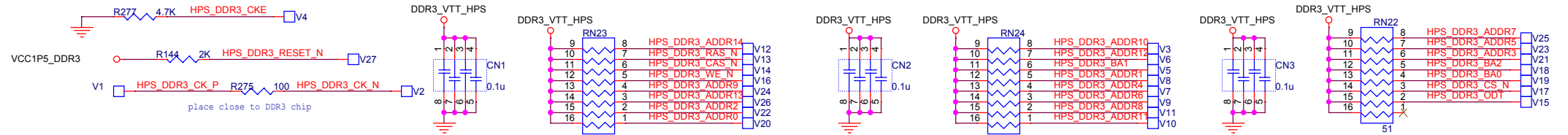




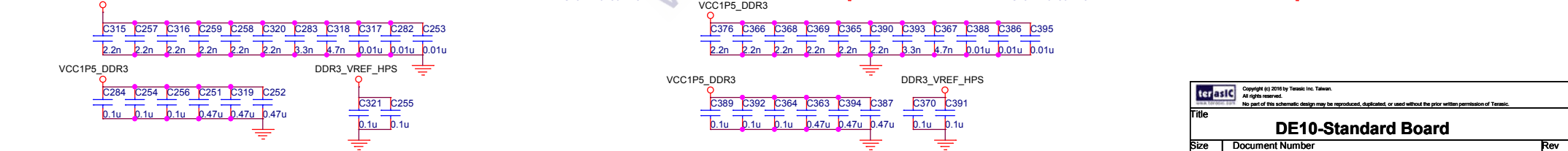
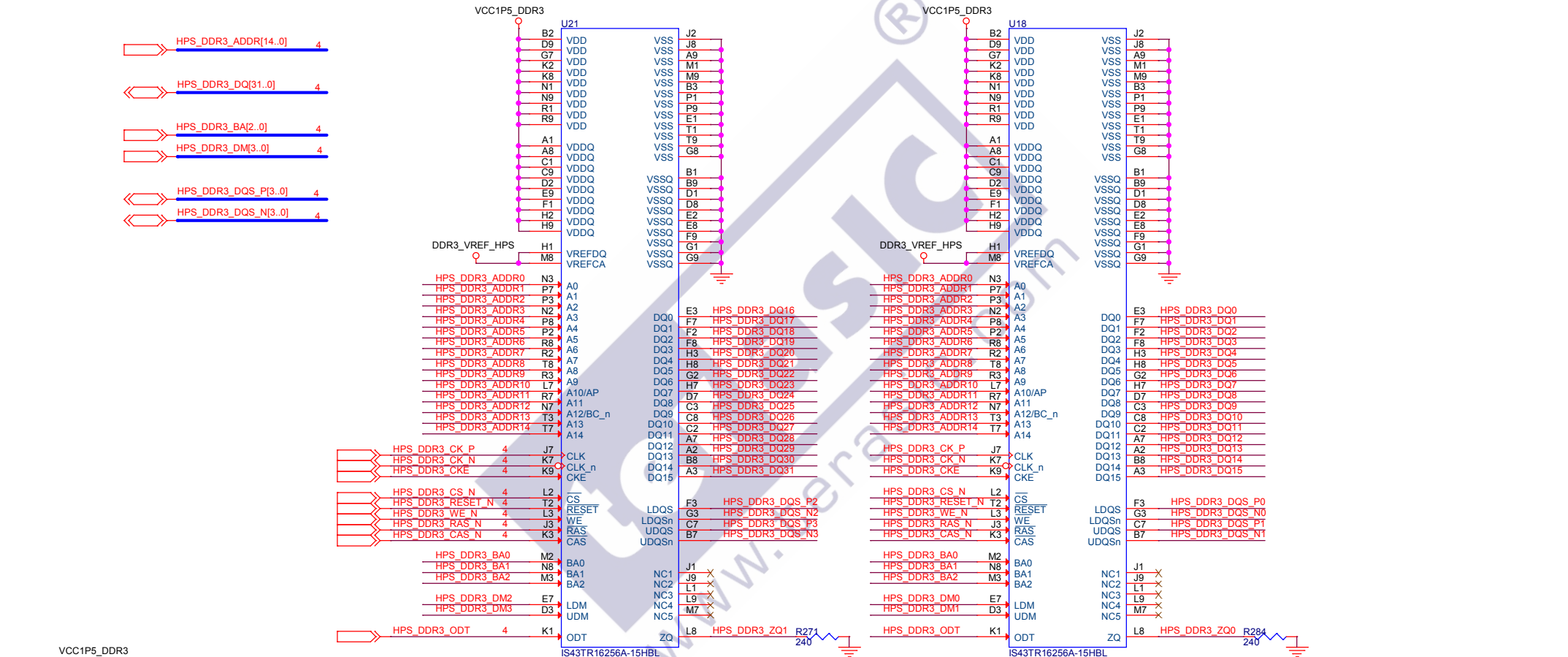
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Title		
DE10-Standard Board		
Size	Document Number	Rev
B	HSMC	D
Date:	Thursday, November 03, 2022	Sheet 14 of 33

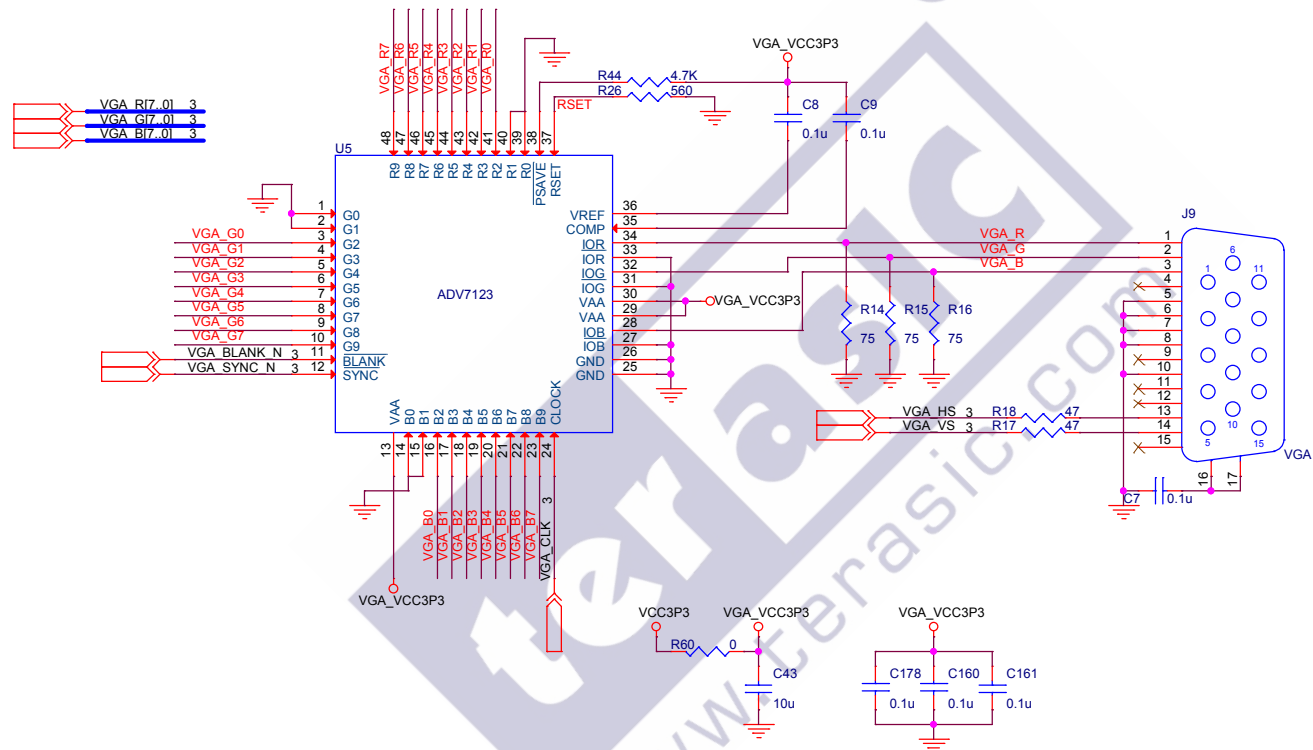


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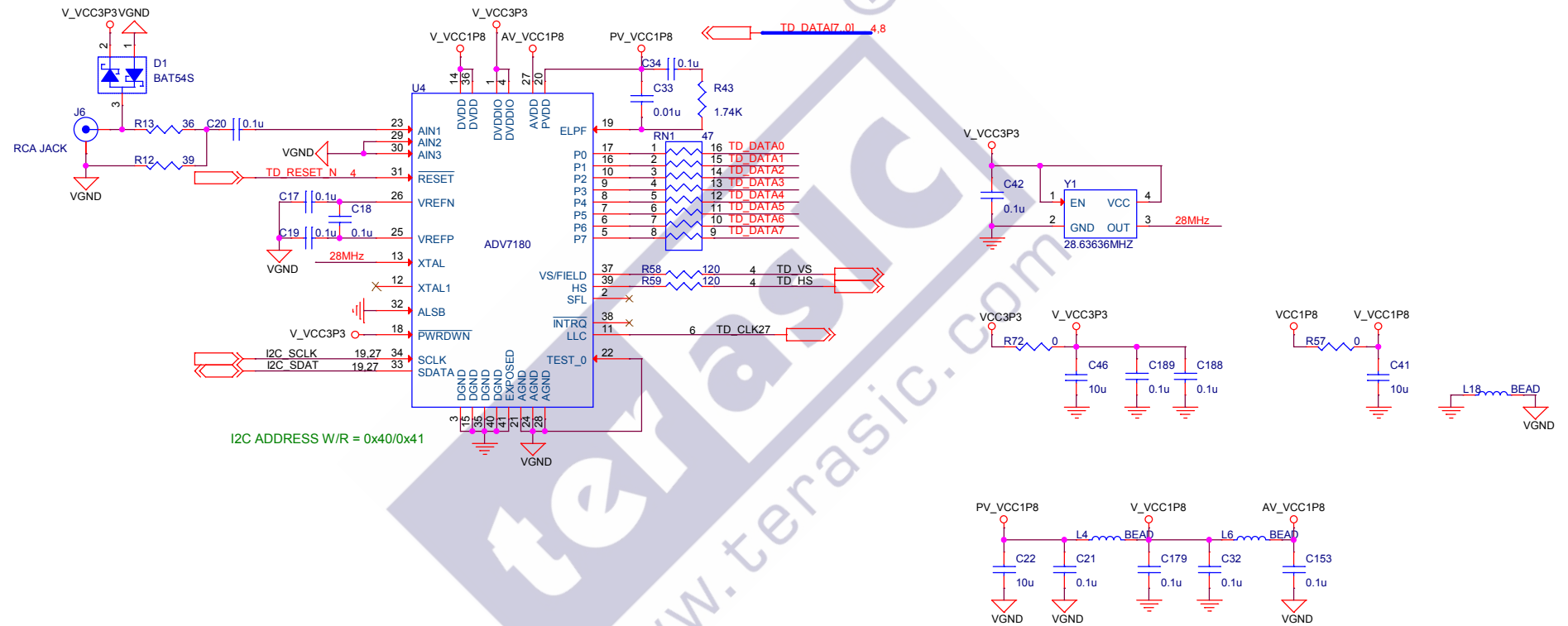


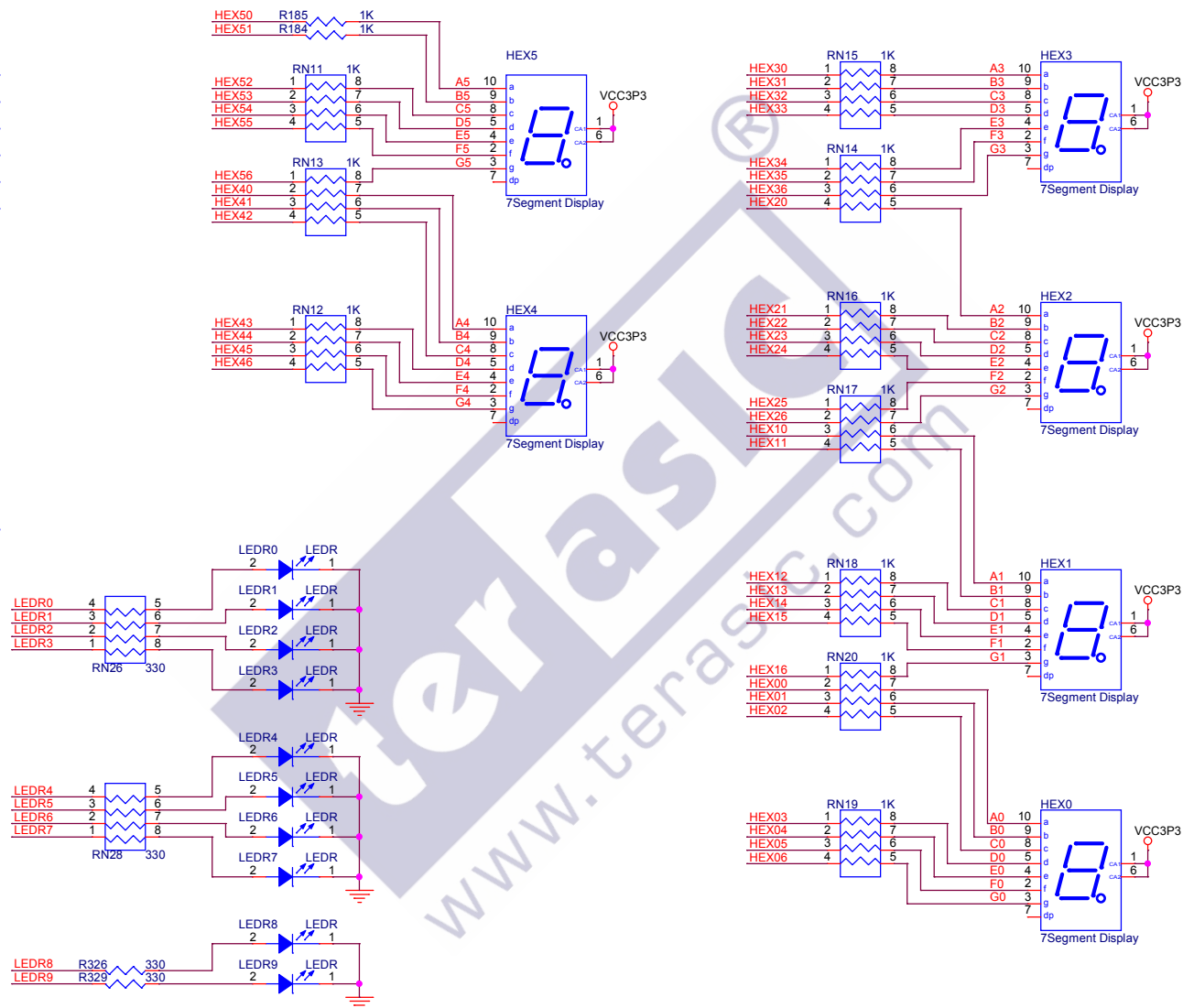
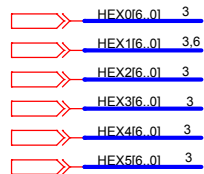
Note: you can swap the signals on the OCT resistor array(include NC pin)

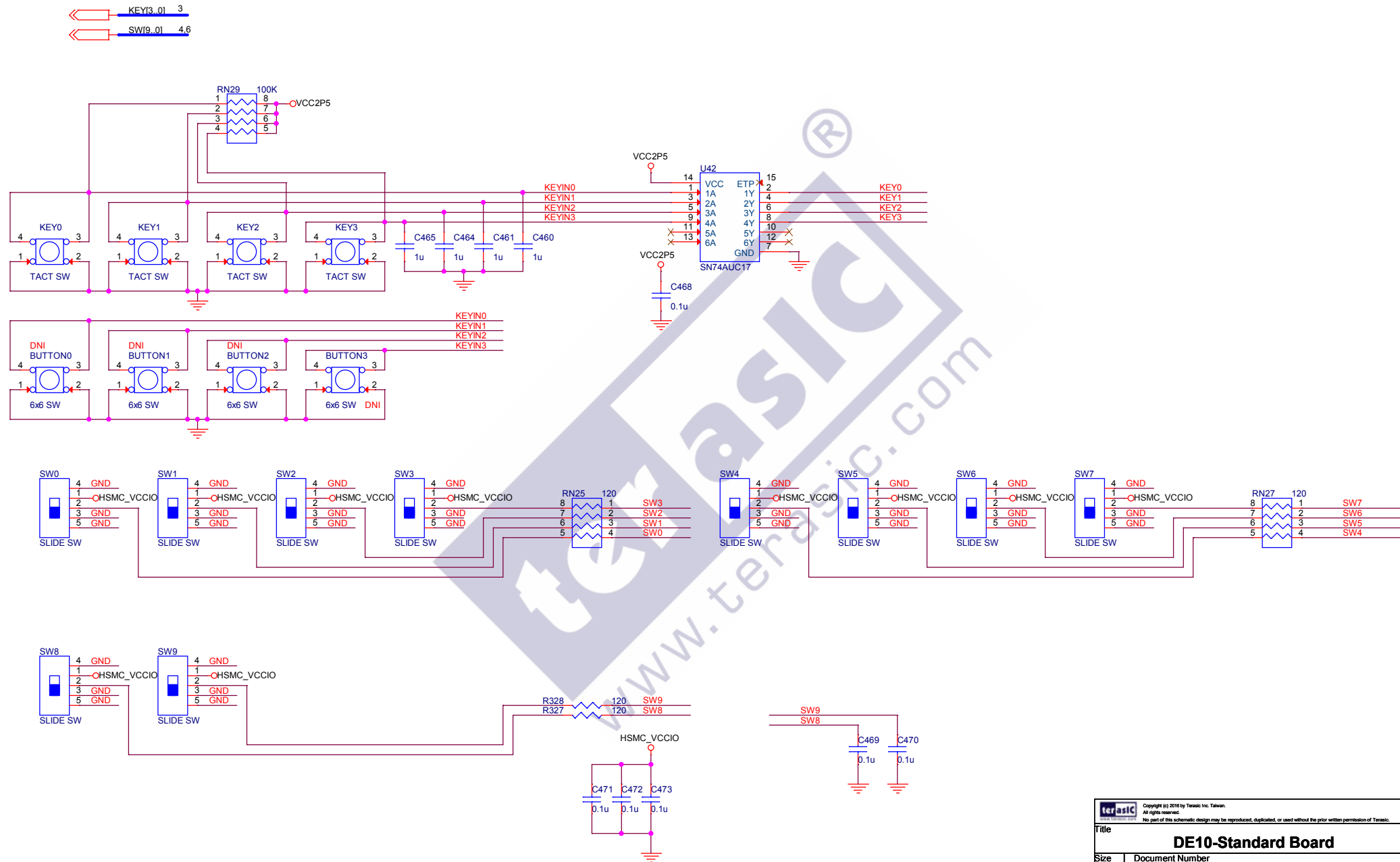


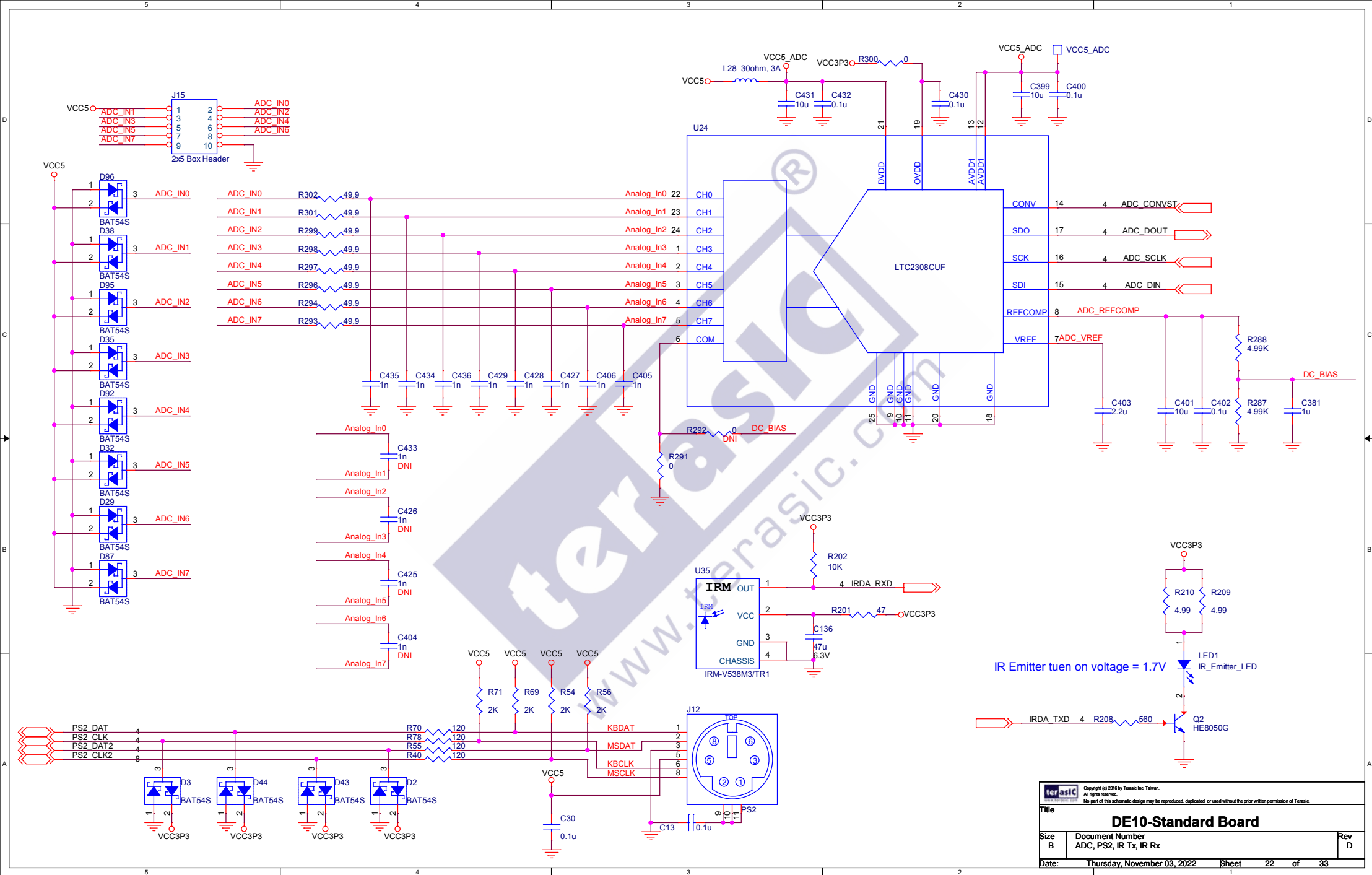


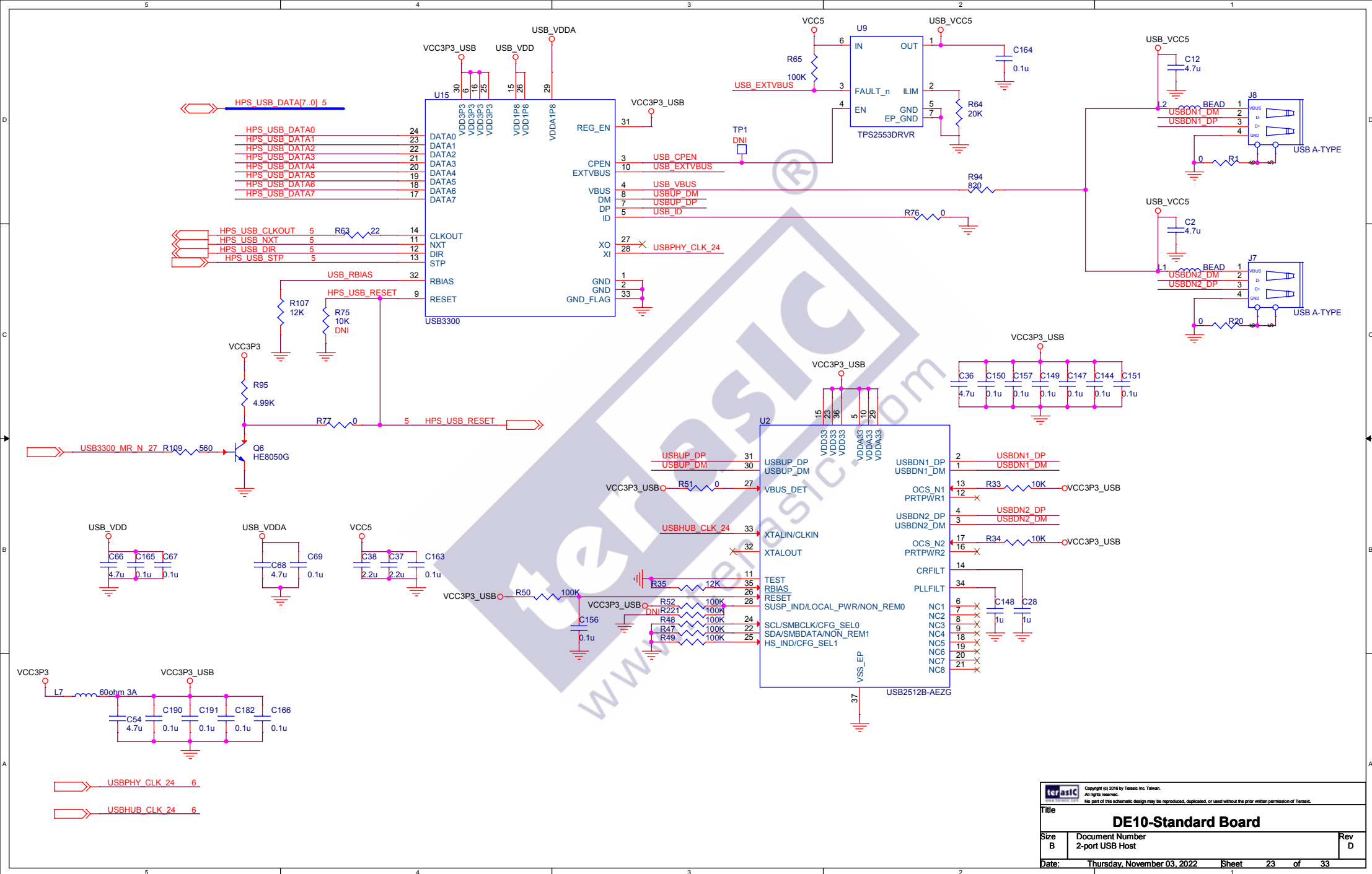
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Title		
DE10-Standard Board		
Size	Document Number	Rev
B	ADV7123 VGA	D
Date:	Thursday, November 03, 2022	Sheet 17 of 33

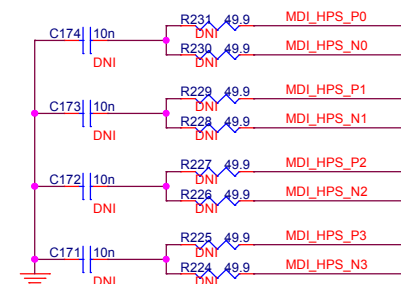
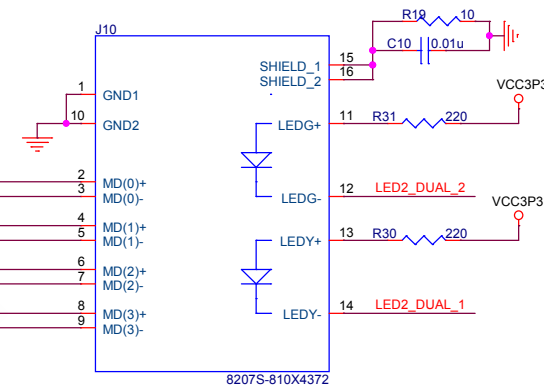
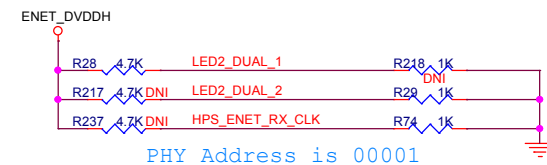
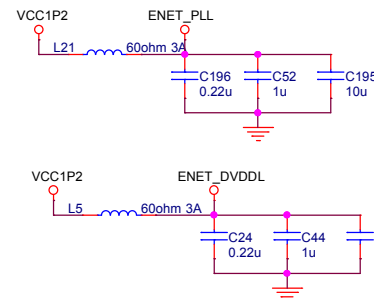
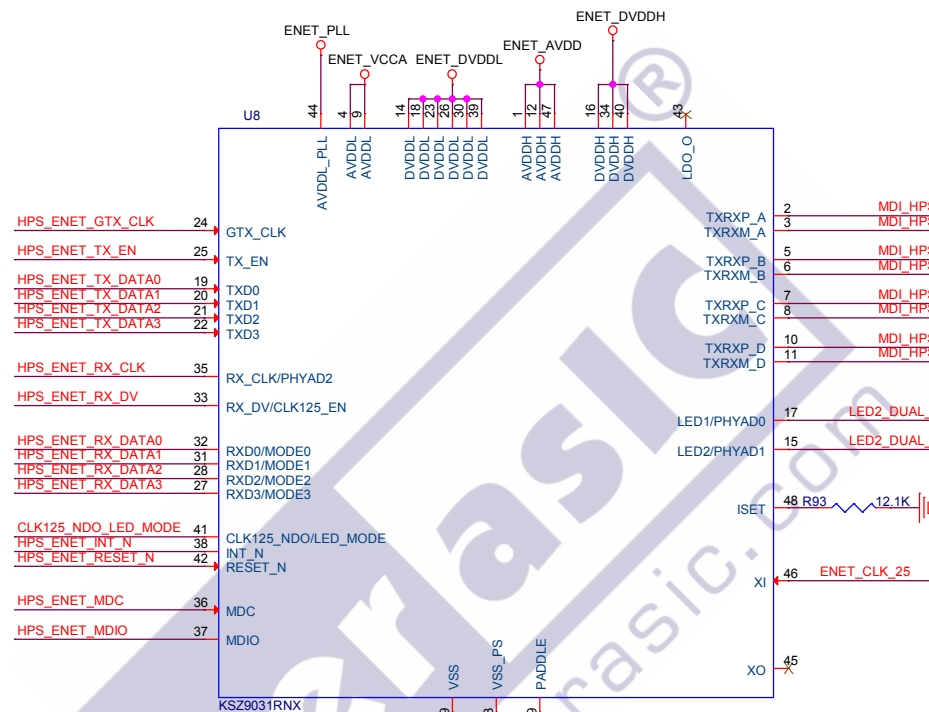
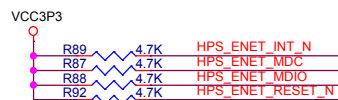
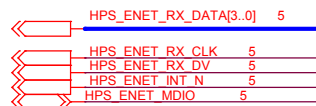


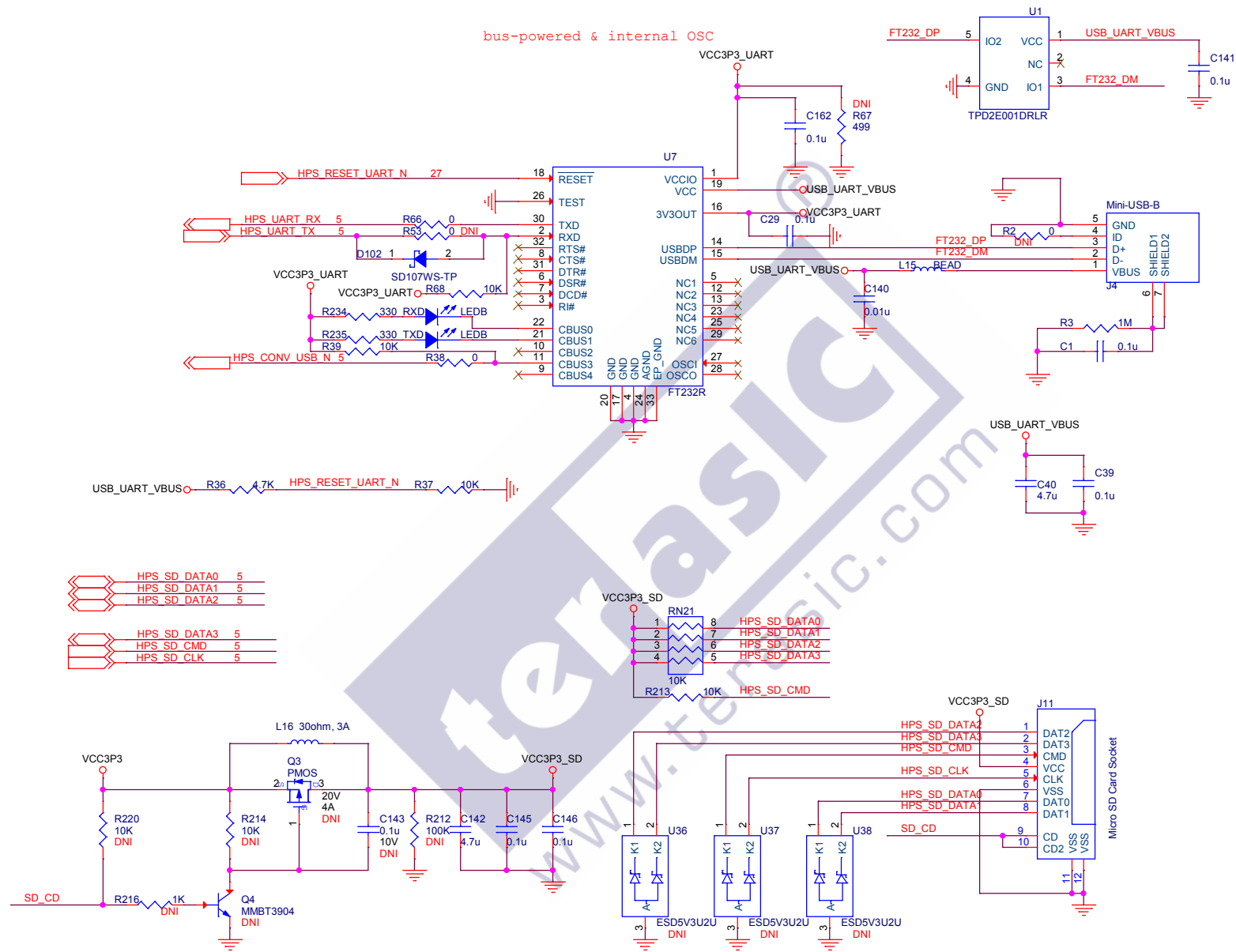










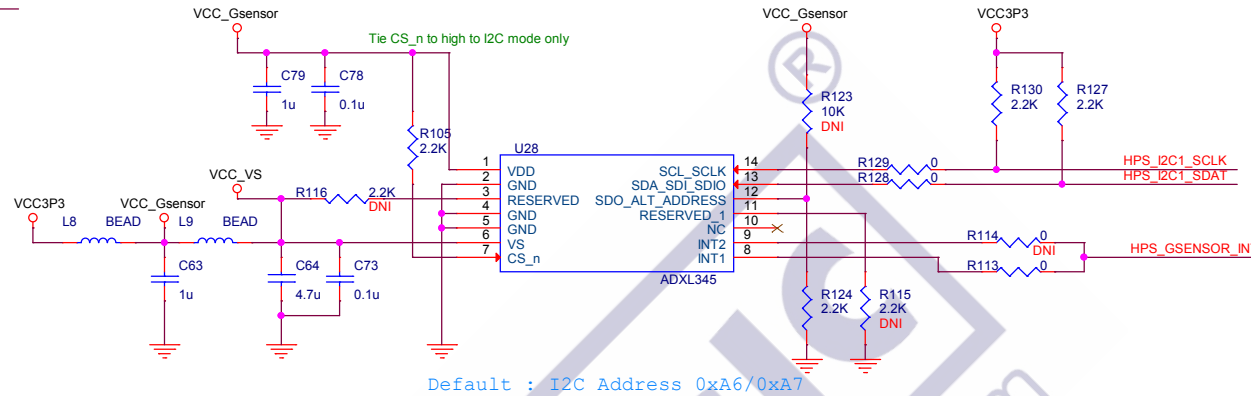


HPS_I2C1_SDAT 5,27

HPS_I2C1_SCLK 5,27

HPS_GSENSOR_INT 5

Digital Accelerometer



HPS_I2C2_SCLK 5

HPS_I2C2_SDAT 5

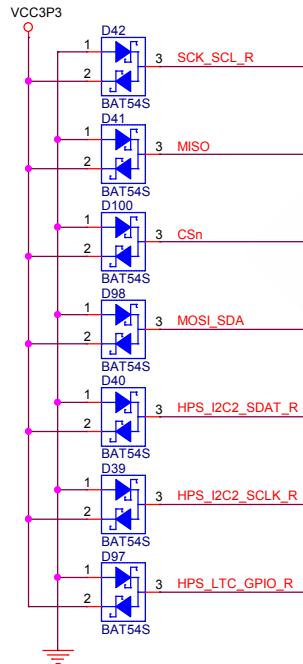
HPS_SPIM_MOSI 5

HPS_SPIM_CLK 5

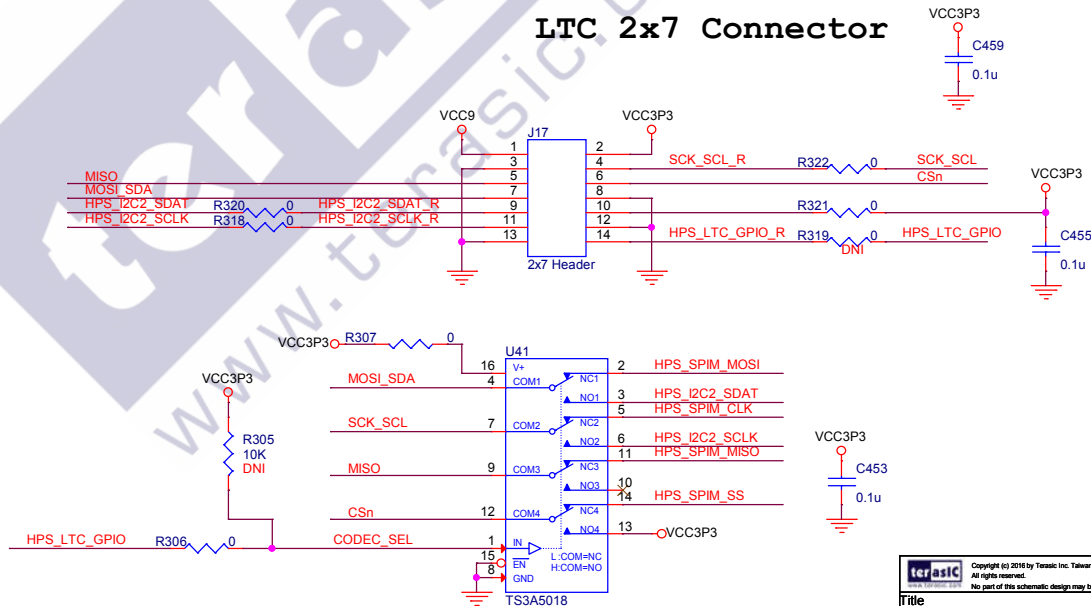
HPS_SPIM_SS 5

HPS_SPIM_MISO 5

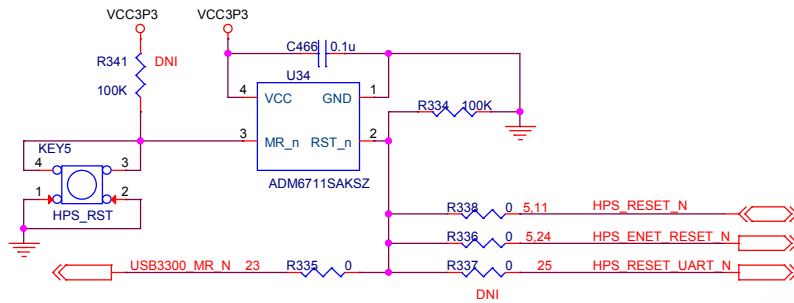
HPS_LTC_GPIO 5



LTC 2x7 Connector

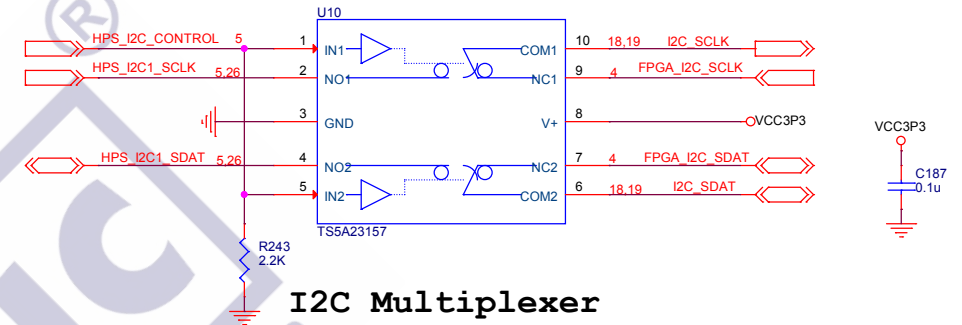


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Title		
DE10-Standard Board		
Size	Document Number	Rev
B	Accelerometer, LTC Connector	D
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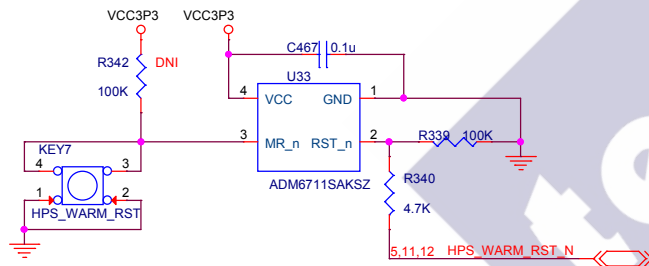


HPS Cold Reset

LOW --> NC to/from COM = ON and NO to/from COM = OFF
HIGH --> NC to/from COM = OFF and NO to/from COM = ON



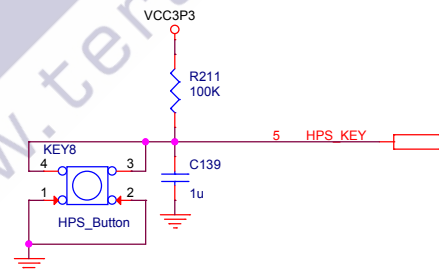
I2C Multiplexer



HPS Warm Reset

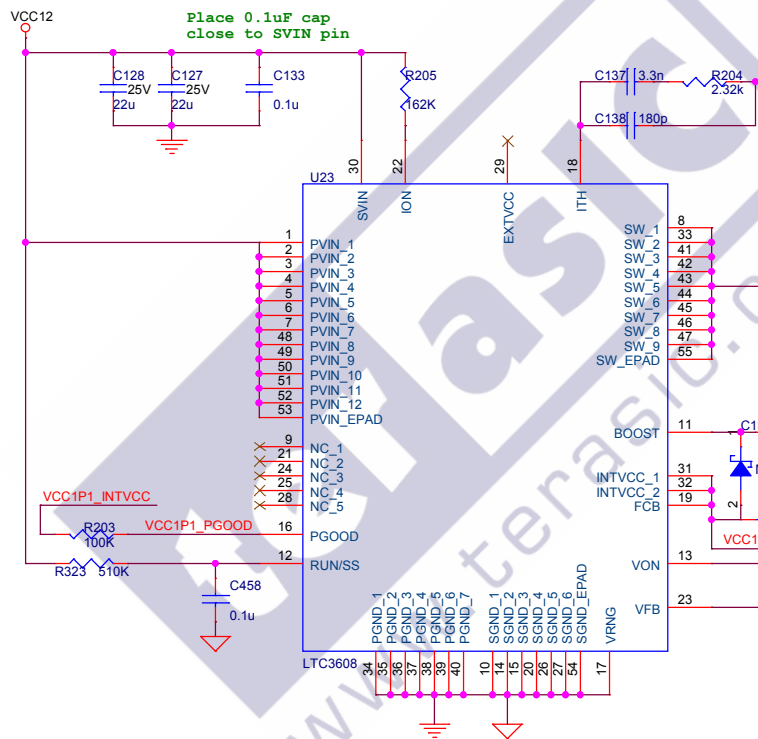
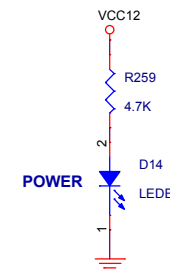
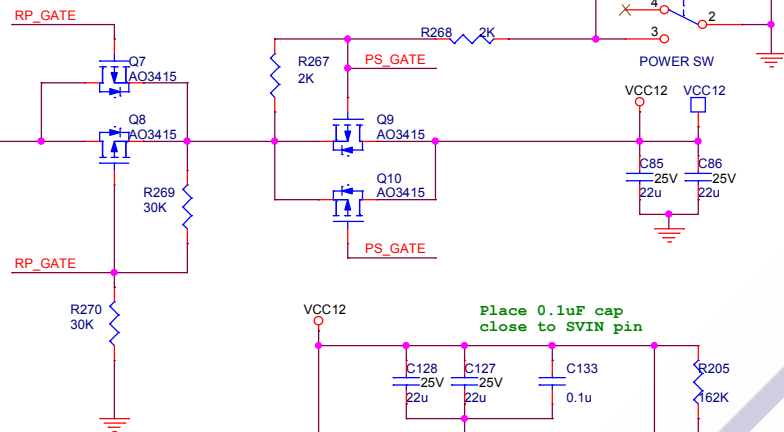


HPS User LED

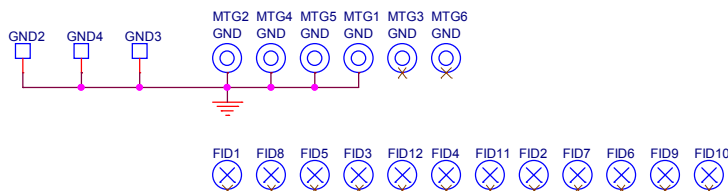
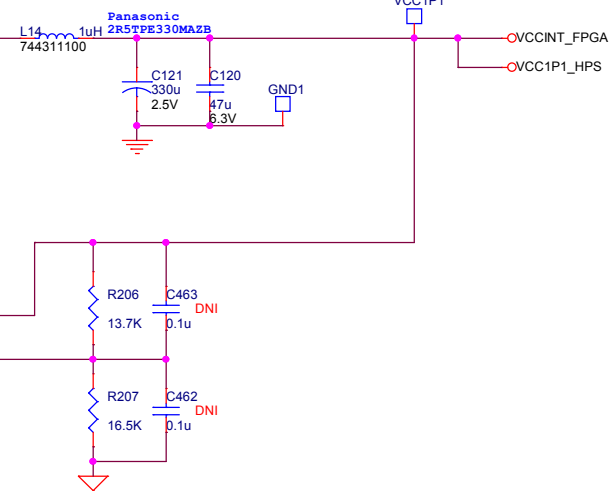


HPS User Button

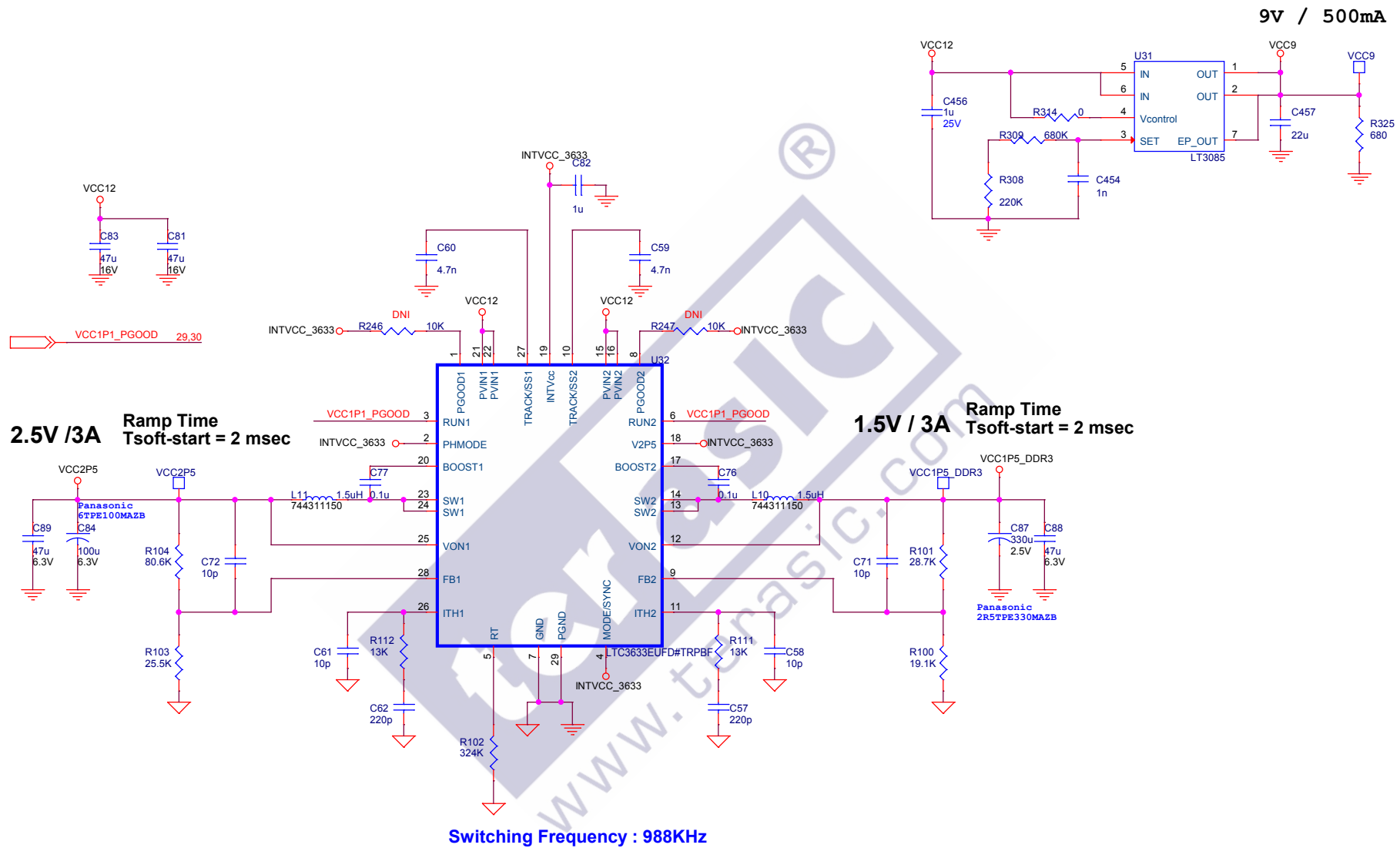
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Title		
DE10-Standard Board		
Size	Document Number	Rev
B	I2C Multiplexer, HPS BUTTON, HPS LED	D
Date:	Thursday, November 03, 2022	Sheet 27 of 33



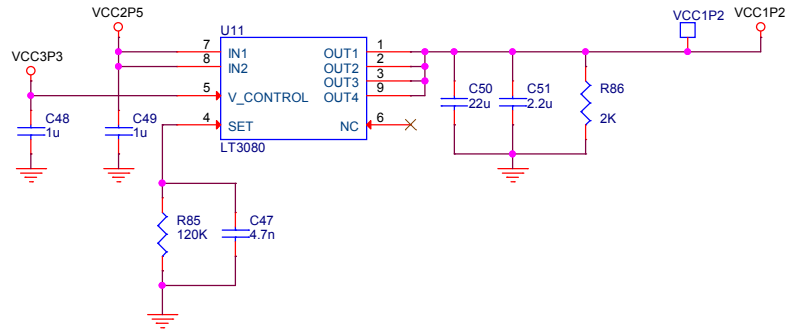
1.1V / 8A
Ramp Time = 190 usec
Switching Frequency : 617KHz



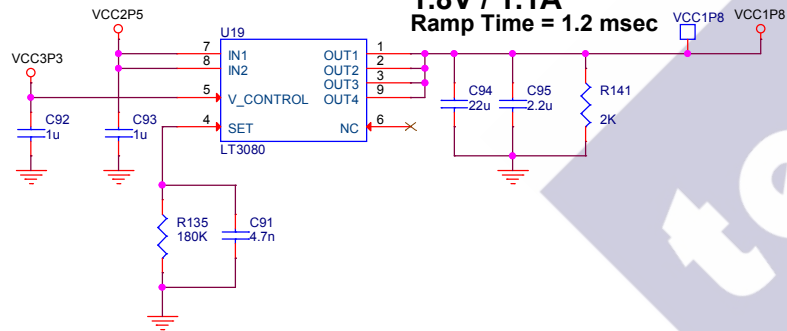
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Title		
DE10-Standard Board		
Size	Document Number	Rev
B	Power - 1.1V	D
Date:	Thursday, November 03, 2022	Sheet 29 of 33



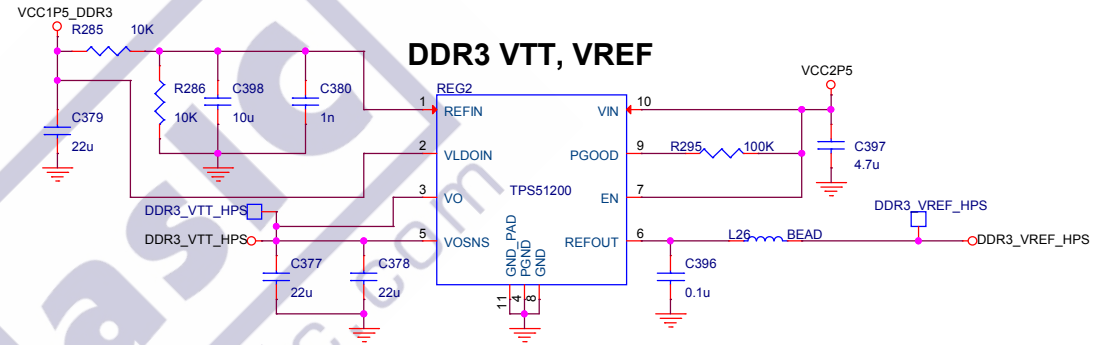
1.2V / 1.1A
Ramp Time = 0.8msec



1.8V / 1.1A
Ramp Time = 1.2 msec

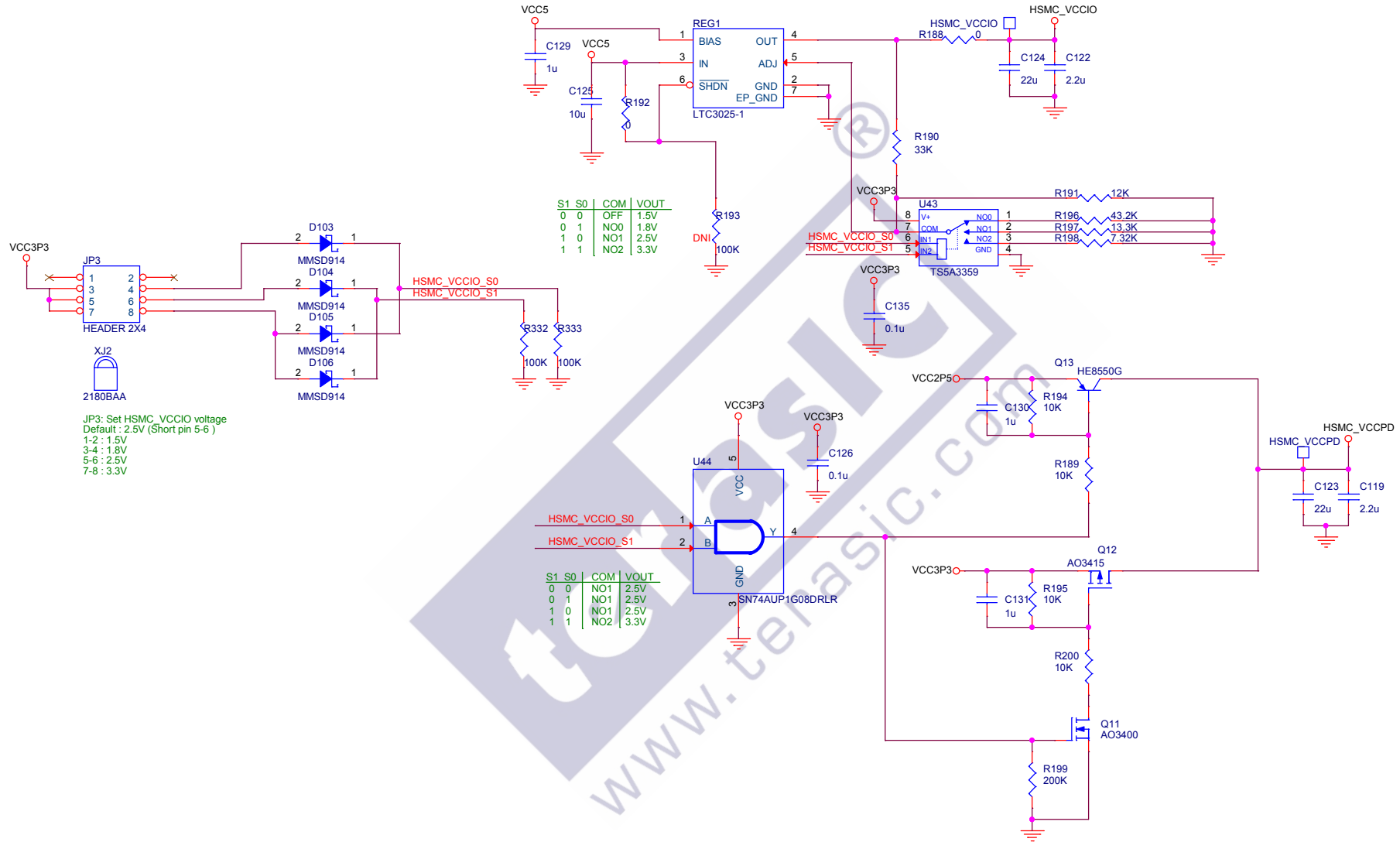


DDR3 VTT, VREF



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Title			
DE10-Standard Board			
Size	Document Number		Rev
B	Power - 1.2V, 1.8V, DDR3 VREF, DDR3 VTT		D
Date:	Thursday, November 03, 2022	Sheet	32 of 33

HSMC_VCCIO / 0.5A Adjustable: 3.3/2.5/1.8/1.5 V
Default: 2.5V



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Title		
DE10-Standard Board		
Size	Document Number	Rev
B	Power - VCCIO_HSMC & HSMC_VCCPD	D
Date:	Thursday, November 03, 2022	Sheet 33 of 33